Warm-start sampling with square-root condition-number dependence

Abstract

For every fixed $\eta > 0$, fixed total variation accuracy, and polynomial squared-L^2 warmness, we construct a finite sampler with an every-execution $C\sqrt{\kappa}\, d^{\eta}$ original-gradient query bound. The assumptions are $\kappa^{-1} I \preceq \nabla^2 V \preceq I$, $\kappa \leqslant d^{100}$, and sufficiently large dimension. A Gaussian conditional probability flow, coupled to a retained momentum and a heat channel, supplies the accelerated transition. The proof uses posterior Poincaré, total covariance, a Poisson-equation modified energy, and one stationary path-entropy comparison. Its finite implementation uses one recursive conditional-mean routine and one simultaneous affine iteration. A single second-moment adjoint calculus controls all derivatives. One order schedule also gives a cubic sufficient threshold and an all-accuracy overhead with exponent 3/4.

Preliminary draft. The arguments and exposition remain subject to further scrutiny. No complete independent verification of the proofs is claimed; corrections and suggestions are welcome.

Source and scope. The Gaussian conditional means, probability transport, Stein centering construction, and recursive original-gradient implementation adapt OpenAI [Ope26]. The present organization uses a conditional covariance invariant, a retained Gaussian input, simultaneous affine evaluation, and a kinetic heat cycle to obtain the warm-start $\sqrt{\kappa}$ dependence. Gaussian conditioning and reverse heat are the proximal-sampling language of [CCSW22, KV26]; the mixing proof uses the Poisson and Bochner toolkit of [Che26, CCLZ26a]. The warm sample is supplied. All guarantees concern gradient-query complexity in exact real arithmetic. Further provenance, adaptation notices, and formalization acknowledgments appear in the final provenance section.

Comparison with Proximal BPS. Chen, Chewi, Lu, and Zhang [CCLZ26a, Corollary 4.4] obtain expected gradient complexity $\widetilde{O}(\sqrt{\kappa}\, d^{1/4})$ from bounded order-2 Rényi divergence, with polylogarithmic accuracy dependence; their explicit bound also allows a general divergence budget, including $O(\log d)$, corresponding to polynomial squared-L^2 warmness. Our fixed-accuracy theorem replaces the dimension power by any fixed d^{η} and gives an every-execution cap, but requires $\kappa \leqslant d^{100}$ and sufficiently large dimension depending on the chosen exponent, warmness, and accuracy. Thus it is not a uniform improvement in all parameter regimes, nor a claim that accelerated warm-start sampling is new. The present guarantees count only original-gradient queries in exact real arithmetic; the all-accuracy statement below should not be confused with polylogarithmic accuracy complexity. In both comparisons the warm input is supplied.

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1 Main results

Write $\pi_F(dx) \propto e^{-F(x)}dx$, $V_t = -\log(e^{-V} * \mathcal{N}(0, tI))$, and $\gamma_d = \mathcal{N}(0, I_d)$. The warmness of ν relative to μ is $B(\nu \mid \mu) = \int (d\nu/d\mu)^2 d\mu = e^{\mathsf{R}_2(\nu\|\mu)}$. All query bounds hold on every execution. Gaussian draws and finite exact-real arithmetic are free; the only oracle is the original gradient.

Theorem 1.1 (Warm-start original-gradient complexity)**.** *Fix $\eta > 0$, $C_w > 0$, and $0 < \epsilon < 1/4$. There are C, d_0, depending only on these choices, such that the following holds. Let $d \geqslant d_0$, $1 \leqslant \kappa \leqslant d^{100}$, and let $V \in C^2(\mathbb{R}^d)$ satisfy $\kappa^{-1} I \preceq \nabla^2 V \preceq I$. Choose a sufficiently small fixed $e = e(\eta) > 0$, put $h = d^{-e}$, and suppose a sample (X_0, p_0) is supplied with*

$$B(\mathsf{Law}(X_0, p_0) \mid \pi_{V_{2h^2}} \otimes \gamma_d) \leqslant d^{C_w}.$$

A finite positive randomized algorithm returns a sample within TV ϵ of π_V, using at most $C\sqrt{\kappa}\, d^{\eta}$ queries to ∇V on every execution. All orders and caps are chosen before the oracle answers. The supplied warm sample is not constructed here.

Corollary 1.2 (Original-target warm input)**.** *The same conclusion holds for a supplied A_0 with $B(\mathsf{Law}(A_0) \mid \pi_V) \leqslant d^{C_w}$. Initialize $X_0 = A_0 + \sqrt{2}hZ_0$ and independent $p_0 \sim \gamma_d$.*

Proof. Forward heat and adjoining an independent Gaussian preserve reference laws and contract order-2 Rényi divergence. □

Theorem 1.3 (A uniform accuracy window and an all-accuracy bound)**.** *Suppose the curvature and condition-number assumptions above hold and an original-target warm input is supplied with warmness at most d^{C_w}, where $C_w \geqslant 0$. For $0 < e \leqslant 1/16$, set*

$$S = (C_w + 1)/2, \qquad K = \left\lceil 10 + \frac{101 + S}{e} \right\rceil, \qquad X_3 = \frac{C_{\mathrm{th}}}{e} K^3 \log(K + 2).$$

There is a source-bounded absolute C_{th} such that, whenever $\log d \geqslant X_3$ and $d^{-1} \leqslant \epsilon < 1/4$, the sampler below has TV error at most ϵ and query cap $C\sqrt{\kappa}\, d^{2e}$. The same window bound holds directly for the smoothed warm input of theorem 1.1. For arbitrary $0 < \epsilon < 1/4$, put $L_0 = 2 + \log(d+1) + \log(1/\epsilon)$. From the original-target warm input, the sampler gives error at most ϵ with

$$Q \leqslant C_{C_w} \sqrt{\kappa}\, \exp\left(C_{C_w} L_0^{3/4} (\log L_0)^{1/4} \right). \tag{1.1}$$

The universal dimension constant is existential; this is an exact-real query bound, not an arithmetic-runtime bound.

Proof. The common order and query certificate in section 13 proves the window bound, the fixed-η specialization, and the all-accuracy bound from the error and work estimates below. □

Proof architecture. The proof uses Gaussian conditioning and conditional variance to build one sampling routine with a retained momentum input. Its probability flow and a Gaussian rotation define the exact accelerated kernel. A Poisson corrector and a modified L^2 energy prove mixing; Gaussian reverse transport and one pathwise KL chain rule transfer mixing to the finite original-gradient implementation. The same noisy sampler then reverses the remaining heat in dyadic steps. A single order schedule pays all of these calls.

2 The sampler

Put $a = 2h^2$, $U = V_a$, $\mu = \pi_U \otimes \gamma_d$, $\ell = \sqrt{a}$, $s = 1/2$, $n = \sqrt{3}/2$, and $\Gamma = h/\sqrt{\kappa + a}$. The supplied smoothed warm state is used directly; original-target input uses corollary 1.2. Each of the N identical phases is the following finite program:

1. Draw $E_0 \sim \gamma_d$ and set $Y_0 = sp + nE_0$, $H_0 = np - sE_0$. Independently draw $\rho \sim \mathrm{Unif}[0, 1/2]$ and put $b = s\rho$. Using this initial split and fresh auxiliary Gaussians, execute $\widehat{W} = \widehat{W}_\rho(X, p; \zeta)$ of proposition 12.6.

2. Draw fresh independent $G_X, G_p \sim \gamma_d$ and set

$$X^+ = X + \ell b(\widehat{W} + G_X), \qquad p^+ = e^{-\Gamma}\widehat{W} + \sqrt{1 - e^{-2\Gamma}}\, G_p. \tag{2.1}$$

For clarity, its final Gaussian assembly is conditional on the fixed initial pair (Y_0, H_0). With $F(z) = V(\sqrt{h}z)$, $r_F = \sqrt{a/h}$, and positive weights $\sum_i \omega_i = \rho$, put $c_i = -sr_F\omega_i$, $\sigma_f = n/(sr_F)$, and $d_i = -\omega_i$. Draw an independent standard Gaussian array Z' and set

$$(p_i)_i = d \otimes H_0 + (I - dd^{\mathsf{T}})^{1/2} Z', \qquad \widehat{W} = p + \sum_i c_i \widehat{g}_i.$$

Here $\widehat{g}_i$ is the finite conditional-gradient correction with supplied input p_i and width σ_f. The remaining Gaussian variance is $\beta^2 = n^2(1 - \sum_i \omega_i^2) \geqslant 9/16$. This is a local conditional Gaussian construction, with one root per phase.

After the phases, discard momentum and perform L cooling steps. At current heat variance v, starting from $v = a$, draw the noisy sampling output $\zeta = \mathsf{S}_1^K(X/\sqrt{h}, \sqrt{v/h})$ for the same F, set $X \leftarrow X + \sqrt{v}\,\zeta/2$, and replace v by $v/2$. Return the last center. To identify this channel, write $R_v^V(x, dz) \propto e^{-V(z) - |z-x|^2/(2v)} dz$. Its exact output is

$$C_v(x) = (x + A)/2 + \sqrt{v}\, G/2, \qquad A \sim R_v^V(x), \quad G \sim \gamma_d \text{ independent.}$$

If $X \sim \pi_{V_v}$, Gaussian conditioning gives $C_v(X) \sim \pi_{V_{v/2}}$. All shapes, orders, and query caps are predetermined and hold at every supplied center.

The exact reference and its invariant law. The retained coupling in proposition 12.6 compares the root with $W = sY_\rho + nH_1$, where Y follows the exact probability flow and, conditional on that path, $H_0 \mapsto H_1$ preserves γ_d. The flow and its common initial split are explicit in equation (5.8) and proposition 5.2. Its marginal law at fixed X is $W \sim \mathsf{Law}(bZ + \sqrt{1 - b^2}G)$, where $X + \ell Z \sim R_a^V(X)$ and $G \sim \gamma_d$ independently. Under the stationary augmentation $X = A + \ell N$, $A \sim \pi_V$, this says

$$W = -bN + \sqrt{1 - b^2}G, \qquad X^+ = A + \ell\{(1 - b^2)N + b\sqrt{1 - b^2}G + bG_X\}.$$

The Gaussian combination in braces is standard and orthogonal to W. Thus $(X^+, W) \sim \mu$; the final momentum OU also preserves μ.

3 One mixing bound and one error bound

The moving Poisson energy in proposition 5.2 gives $c\|f\|_2^2 \leqslant \mathcal{E}(f) \leqslant C\|f\|_2^2$ and $\mathcal{E}(P^*f) \leqslant (1 + c\Gamma)^{-1}\mathcal{E}(f)$ for centered f. Its dormant-noise estimate follows from equation (12.22). Cooling contracts TV, so lemma 4.2 gives, with $B \leqslant d^{C_w}$,

$$\mathsf{TV}(\mathsf{Law}(X_{\mathrm{out}}^{\mathrm{exact}}), \pi_V) \leqslant Ce^{-cN\Gamma}\sqrt{B - 1} + \tfrac{1}{2}\sqrt{da2^{-L}}. \tag{3.1}$$

The single quantitative interface. For $y = \log(1/h) \geqslant CK^3 \log(K+2)$, the dimension-free certificate in section 13 supplies, under stationary input, root RMS error $C\sqrt{d}\, h^{K-1}$ and cap $h^{-1/32}$. To compare a phase, first expose

$$(W + G_X,\ e^{-\Gamma} W + \sqrt{1 - e^{-2\Gamma}} G_p)$$

and then map its first coordinate to $X + \ell b(W + G_X)$. The fresh Gaussian reserves give reverse KL at most Cdh^{2K-2}/Γ; deterministic data processing preserves this bound uniformly in the clock ρ. Each cooling call has stationary reverse KL at most Cdh^{2K-2} and the same query cap, by proposition 12.5. All comparisons retain their entering states. The warm path bound lemma 4.4 consequently gives

$$\mathsf{TV}(\mathsf{Law}(X_{\mathrm{out}}), \pi_V) \leqslant Ce^{-cN\Gamma}\sqrt{B-1} + \tfrac{1}{2}\sqrt{da2^{-L}} + C\sqrt{Bd\, h^{2K-2}(N/\Gamma + L)}. \tag{3.2}$$

The complete deterministic query cap is

$$Q \leqslant C(N+L)h^{-1/32}. \tag{3.3}$$

The single order certificate chooses N, L, K, h and pays these two bounds, proving all three public conclusions with the stated cubic threshold and all-accuracy exponent $3/4$.

4 Gaussian conditioning and one entropy account

Write $\pi_t = \pi_V * \mathcal{N}(0, tI)$, $V_t = -\log \pi_t$, and $R^F_{t,y}(\mathrm{d}z) \propto e^{-F(z) - |z-y|^2/(2t)} \mathrm{d}z$. These are comparison laws, not additional oracles. Gaussian augmentation and its posterior reverse are the channels underlying the proximal sampler and restricted-Gaussian-oracle framework of Lee, Shen, and Tian [LST21]; see [CCSW22, KV26] for subsequent analyses.

Lemma 4.1 (Posterior score calculus)**.** *Let $F \in C^2$ satisfy $mI \preceq \nabla^2 F \preceq \lambda I$ with $m \geqslant 0$. Put $F_t = -\log(e^{-F} * \mathcal{N}(0, tI))$ and $A \sim R^F_{t,y}$. Then*

$$\nabla F_t(y) = \mathbb{E}\nabla F(A) = t^{-1}(y - \mathbb{E}A), \tag{4.1}$$

$$\nabla^2 F_t(y) = \mathbb{E}\nabla^2 F(A) - \mathrm{Cov}(\nabla F(A)) = t^{-1} I - t^{-2}\,\mathrm{Cov}(A), \tag{4.2}$$

$$\frac{m}{1+mt} I \preceq \nabla^2 F_t \preceq \lambda I. \tag{4.3}$$

When $t\lambda < 1$, the same posterior satisfies

$$\mathrm{Cov}(\nabla F(A)) \preceq \frac{t\lambda}{1 - t\lambda} \nabla^2 F_t(y). \tag{4.4}$$

For the primitive law $Z \sim \nu_{x,r}$ and $Y = \rho Z + \sqrt{1-\rho^2} G$, $G \sim \gamma_d$ independent, put $t = r^2(1-\rho^2)$ and $c = x + r\rho Y$. Then

$$\mathsf{Law}(x + rZ \mid Y) = R^F_{t,c}, \qquad M_\rho(y) = \nabla F_t(x + r\rho y), \qquad H_\rho(y) = \tfrac{1}{2}|y|^2 + F_t(x + r\rho y) + \text{constant}.$$

Proof. Differentiate the Gaussian convolution through either factor and integrate by parts to obtain the score and Hessian identities. The posterior is $(m + t^{-1})$-strongly log-concave; its Poincaré inequality gives $\mathrm{Cov}(A) \preceq t(1+mt)^{-1} I$, proving the lower curvature bound. The first Hessian identity proves the upper bound. For $\Sigma = \mathrm{Cov}(\nabla F(A))$, posterior Poincaré and $0 \preceq \nabla^2 F \preceq \lambda I$ give

$$\Sigma \preceq t\mathbb{E}(\nabla^2 F(A))^2 \preceq t\lambda \mathbb{E}\nabla^2 F(A) = t\lambda(\nabla^2 F_t + \Sigma),$$

which proves (4.4). Finally complete the square in $e^{-F(x+rz)}\phi_1(z)\phi_{1-\rho^2}(y - \rho z)$. Gaussian posterior tails justify all differentiations for C^2 inputs. □

Lemma 4.2 (Gaussian heat channels)**.** *For $A \sim \pi_V$ and independent standard Gaussians, put $X = A + \sqrt{a}Z_1$ and $Y = X + \sqrt{b}Z_2$. Then*

$$X \sim \pi_a, \quad Y \sim \pi_{a+b}, \quad A \mid Y \sim R^V_{a+b,Y}, \quad X \mid (A,Y) \sim \mathcal{N}\left(\frac{bA+aY}{a+b}, \frac{ab}{a+b}I\right).$$

Every conditional channel contracts relative-density L^2 norms between its reference marginals. In particular, the reverse heat bridge

$$D_w(y) = \tfrac{1}{2}(y + A + \sqrt{w}G), \qquad A \sim R^V_{w,y}, \quad G \sim \gamma_d \text{ independent},$$

carries π_w to $\pi_{w/2}$.

Proof. Complete the square in $e^{-V(A)}\phi_a(X-A)\phi_b(Y-X)$. The output relative density is the conditional expectation of the input relative density; conditional Jensen proves the contraction. The final statement is the equal-width case $a = b = w/2$. □

Lemma 4.3 (Gaussian reverse transport)**.** *For any coupling $(U, \widehat{U})$ and positive definite Σ,*

$$\mathsf{KL}\big(\mathsf{Law}(U+G) \| \mathsf{Law}(\widehat{U}+G)\big) \leqslant \tfrac{1}{2}\mathbb{E}|U-\widehat{U}|^2_{\Sigma^{-1}}, \qquad G \sim \mathcal{N}(0,\Sigma)$$

when G is independent of the coupling. The same statement holds conditionally on any retained input.

Proof. Condition on the coupling, use the relative entropy between Gaussian shifts, and then apply joint convexity. This is the Gaussian regularization inequality used in [CCLZ26b, Lemma 6.3(i)]. □

Lemma 4.4 (Warm path comparison)**.** *Let exact channels P_j carry reference laws μ_{j-1} to μ_j, and let $\widehat{P}_j$ be their finite implementations. Suppose*

$$\int \mathsf{KL}\big(P_j(z,\cdot) \| \widehat{P}_j(z,\cdot)\big)\, \mu_{j-1}(\mathrm{d}z) \leqslant \delta_j^2.$$

If $\nu \ll \mu_0$ and $\int (\mathrm{d}\nu/\mathrm{d}\mu_0)^2 \mathrm{d}\mu_0 \leqslant B$, the two chains started from ν satisfy

$$\mathsf{TV}(\nu P_1 \cdots P_n, \nu \widehat{P}_1 \cdots \widehat{P}_n) \leqslant \sqrt{\frac{B}{2}\sum_{j=1}^n \delta_j^2}.$$

Proof. Retain the initial state in the visible-state path, and let $\mathsf{P}_z, \widehat{\mathsf{P}}_z$ be the conditional path laws. The KL chain rule, oriented from exact to implemented, gives

$$\int \mathsf{KL}(\mathsf{P}_z \| \widehat{\mathsf{P}}_z)\, \mu_0(\mathrm{d}z) \leqslant \sum_j \delta_j^2.$$

Writing $f = \mathrm{d}\nu/\mathrm{d}\mu_0$, conditional Pinsker and Cauchy–Schwarz give

$$\int f(z)\mathsf{TV}(\mathsf{P}_z, \widehat{\mathsf{P}}_z)\, \mu_0(\mathrm{d}z) \leqslant \|f\|_2 \left[\tfrac{1}{2}\int \mathsf{KL}(\mathsf{P}_z \| \widehat{\mathsf{P}}_z)\, \mu_0(\mathrm{d}z)\right]^{1/2}.$$

Finally discard the path. Only exact reference prefixes enter the entropy sum. This combines the usual KL chain rule [AC24, Eq. (2.1)] with the order-2 Rényi warmness viewpoint of [KV26]. The path does not include oracle transcripts or the compiler's raw tapes. □

The terminal decoder uses the same noisy sampler. For $0 < w \leqslant h$ use the public sampling block with

$$F(z) = V(\sqrt{h}\, z), \qquad x = y/\sqrt{h}, \qquad r = \sqrt{w/h}.$$

Its target is $\zeta = (A - y)/\sqrt{w} + G$ with $A \sim R^V_{w,y}$ and independent $G \sim \gamma_d$. Suppose its pre-reserve Wasserstein error is at most $\varepsilon_S D_F(x)$ and its every-execution cap is Q_S; proposition 12.5 supplies this contract with $D_F(x) = \sqrt{d} + |\nabla F(x)|$ and an independent reserve of covariance $I/4$. Define

$$\widehat{D}_w(y) = y + \tfrac{1}{2}\sqrt{w}\,\widehat{\zeta}. \tag{4.5}$$

The Gaussian reverse-transport lemma gives

$$\int \mathsf{KL}(D_w(y,\cdot) \| \widehat{D}_w(y,\cdot))\, \pi_w(\mathrm{d}y) \leqslant C d \varepsilon_S^2. \tag{4.6}$$

Indeed $Y = A + \sqrt{w}Z$ under π_w, $\mathbb{E}_{\pi_V}|\nabla V|^2 = \mathbb{E}_{\pi_V}\Delta V \leqslant d$, and ∇V is 1-Lipschitz, so $\mathbb{E}D_F(Y/\sqrt{h})^2 \leqslant Cd$. The same moment bound holds for every heat width encountered by the normalized sampling root.

Proposition 4.5 (Dyadic heat decoder)**.** *For $0 < w \leqslant h$, set $w_j = w2^{-j}$ and perform*

$$Y_0 = y, \qquad Y_{j+1} = \widehat{D}_{w_j}(Y_j), \quad 0 \leqslant j < L, \qquad A_{\mathrm{out}} = Y_L.$$

Let H_w be the corresponding composition of exact bridges, and $\widehat{H}_w$ the displayed finite decoder. Then $\pi_w H_w = \pi_{w2^{-L}}$, its cap is LQ_S at every center, and

$$\int \mathsf{KL}(H_w(y,\cdot) \| \widehat{H}_w(y,\cdot))\, \pi_w(\mathrm{d}y) \leqslant C d L \varepsilon_S^2. \tag{4.7}$$

The remaining bias obeys $\mathsf{TV}(\pi_{w2^{-L}}, \pi_V) \leqslant \frac{1}{2}\sqrt{dw2^{-L}}$.

Proof. Sum (4.6) along the exact heat marginals by the KL chain rule and discard intermediate states. For $A \sim \pi_V$, direct density calculation, $\mathbb{E}\nabla V(A) = 0$, and $\nabla^2 V \preceq I$ give

$$\mathsf{KL}(\mathsf{Law}(A + z) \| \pi_V) = \mathbb{E}[V(A + z) - V(A)] \leqslant |z|^2/2.$$

Average over $z \sim \mathcal{N}(0, w2^{-L}I)$ and use convexity and Pinsker. This bias is paid once, without a warmness factor. □

5 Mixing of the Gaussian conditional lift

Put

$$a = 2h^2, \qquad \ell = \sqrt{a}, \qquad \rho_* = 1/2, \qquad s = 1/2, \qquad n = \sqrt{3}/2,$$
$$m = (\kappa + a)^{-1}, \quad \gamma = \sqrt{m}, \quad \Gamma = \gamma h.$$

Write $U_\rho = V_{a(1-\rho^2)}$ and $\mu_\rho(\mathrm{d}x) \propto e^{-U_\rho(x)}\mathrm{d}x$. The posterior curvature bound gives $mI \preceq \nabla^2 U_\rho \preceq I$ for $0 \leqslant \rho \leqslant \rho_*$. The same bounds hold along each forward heat segment used below. We denote the phase position by X_ρ and the Poisson corrector by $\mathcal{A}_\rho$, to distinguish these two operators and states.

5.1 The Poisson corrector and its moving reference

The modified-L^2 strategy originates in Dolbeault, Mouhot, and Schmeiser [DMS15]. The unshifted reduced-inverse corrector used here is the $\alpha = 0$ case of Fan, Li, and Lu [FLL26, Section 2, Eqs. (26)–(29)]. Neither this corrector nor the basic hypocoercive mechanism is new here; the present analysis tracks a moving reference law and a discrete kinetic heat cycle. For the underlying Poisson dualization and Bochner identities, see [Che26, Sections 2.2.1–2.2.2, Eqs. (2.2.4)–(2.2.5) and Lemma 2.2.7]; the closure and moving-reference steps needed here are addressed below. We first work with one smooth potential U satisfying $mI \preceq \nabla^2 U \preceq I$. Let $\mu(\mathrm{d}x) \propto e^{-U(x)}\mathrm{d}x$, $D = \nabla_x$, and

$$L = D^*D = -\Delta + \nabla U \cdot \nabla.$$

The inverse L^{-1} always means the inverse on $L^2_0(\mu) = \{g : \int g \,\mathrm{d}\mu = 0\}$, extended by zero on constants. Poincaré gives $L \succeq mI$ on this space. For $u = L^{-1}g$, two integrations by parts give

$$\|g\|_2^2 = \|Lu\|_2^2 = \sum_i \langle \partial_i u, \partial_i L u\rangle = \|\nabla^2 u\|_{\mathrm{HS},2}^2 + \|(\nabla^2 U)^{1/2}\nabla u\|_2^2. \tag{5.1}$$

Here we used $\partial_i L u = L\partial_i u + \sum_j (\partial_{ij}U)\partial_j u$. The identity first holds on a smooth core and extends to the Poisson solution by closure of the Dirichlet form and the displayed estimate.

On $L^2(\mu \otimes \gamma_d \otimes \gamma_d)$, let Π average the two Gaussian coordinates P, E. Define the isometry

$$J_P v(x,P,E) = P \cdot v(x), \qquad J_P^* f(x) = \int P f(x,P,E)\, \gamma_d(\mathrm{d}P)\gamma_d(\mathrm{d}E).$$

Thus $J_P^*\Pi = 0$. With $T_P = P \cdot \nabla_x - \nabla U \cdot \nabla_P$, define

$$\mathcal{A} = L^{-1}(T_P\Pi)^* = L^{-1}D^*J_P^*, \qquad \mathcal{A}^* = J_P D L^{-1}\Pi.$$

Scalar outputs of $\mathcal{A}$ are identified with functions constant in the momentum coordinates. In particular, $\mathcal{A}\Pi = 0$, $\Pi\mathcal{A} = \mathcal{A}$, and $\mathcal{A}^* f = P \cdot \nabla L^{-1}\Pi f$.

The operator $\mathsf{C} = DL^{-1}D^*$ is the orthogonal projection onto the closure of gradient fields. Indeed, if $u = L^{-1}D^*v$, its weak equation says $\langle v - Du, D\phi\rangle = 0$ for every ϕ; hence Du is precisely that projection. This proves $\|D\mathcal{A}\| \leqslant 1$. Gaussian isometry and (5.1) also give

$$\|D\mathcal{A}^* f\|_2^2 = \|\nabla^2 L^{-1}\Pi f\|_{\mathrm{HS},2}^2 \leqslant \|\Pi f\|_2^2.$$

Finally $\|DL^{-1}\| \leqslant \gamma^{-1}$, by Poincaré or spectral calculus. Consequently, writing $M = \|\Pi f\|_2$, $R = \|(I-\Pi)f\|_2$, we have

$$\|\mathcal{A}\| \leqslant \gamma^{-1}, \qquad \|D\mathcal{A}\|, \|D\mathcal{A}^*\| \leqslant 1, \qquad |\langle \mathcal{A}f, f\rangle| \leqslant \gamma^{-1}MR. \tag{5.2}$$

All these bounds also hold with no hidden coordinate E.

Choose an absolute $0 < \theta \leqslant 1/4$, to be decreased below, and put

$$\varepsilon_0 = \theta\gamma, \qquad J_U(f) = \langle \mathcal{A}f, f\rangle, \qquad \mathcal{E}_U(f) = \tfrac{1}{2}\|f\|_2^2 + \varepsilon_0 J_U(f).$$

The corrector bound gives the explicit norm equivalence

$$\frac{1-\theta}{2}\|f\|_2^2 \leqslant \mathcal{E}_U(f) \leqslant \frac{1+\theta}{2}\|f\|_2^2. \tag{5.3}$$

Thus $\mathcal{E}_U$ is a positive quadratic form, in particular convex.

Pullback to a fixed Hilbert space. Suppose $\mu_t(\mathrm{d}x) \propto e^{-U_t(x)}\mathrm{d}x$ satisfies

$$\partial_t \mu_t + \operatorname{div}(\mu_t v_t) = 0, \qquad v_t = -c_t \nabla U_t, \qquad mI \preceq H_t := \nabla^2 U_t \preceq I.$$

Let Φ_t be the flow of v_t, starting from the identity. The bounded Hessians make this a complete differentiable flow on each compact time interval. It carries μ_0 to μ_t. Composition with Φ_t is therefore a unitary map from $L^2(\mu_t)$ to $L^2(\mu_0)$; for vector fields we compose each Euclidean component, without multiplying by the Jacobian. All adjoints in the following calculation are taken in these fixed scalar and vector Hilbert spaces.

Write D_t for the pulled-back spatial gradient. If $J_t = D\Phi_t$, then

$$D_t f = J_t^{-\mathsf{T}} \nabla f, \qquad \dot{J}_t = -c_t (H_t \circ \Phi_t) J_t.$$

Suppressing the composition in the notation for H_t, differentiation therefore gives

$$\dot{D}_t = c_t H_t D_t, \qquad \dot{D}_t^* = c_t D_t^* H_t, \qquad \dot{L}_t = 2c_t D_t^* H_t D_t, \quad L_t = D_t^* D_t.$$

The second identity follows by differentiating the adjoint relation in the fixed Hilbert spaces; it has no derivative of the measure. The constant function is unchanged by pullback, so the centered subspace and its projection are fixed as well.

For a fixed centered g, let $u_t = R_t g$, $R_t = L_t^{-1}$. Its weak Poisson equation is $\langle D_t u_t, D_t \phi\rangle = \langle g, \phi\rangle$. Differentiating this equation yields

$$\langle D_t \dot{u}_t, D_t \phi\rangle = -2c_t \langle H_t D_t u_t, D_t \phi\rangle, \qquad \dot{R}_t = -2c_t R_t D_t^* H_t D_t R_t.$$

These identities can be obtained directly from difference quotients: subtract the weak equations at two times, use their common centered coercivity m, and pass to the limit. On a compact time interval the pulled-back gradients have equivalent form norms and $\dot{D}_t = c_t H_t D_t$ is bounded relative to that form norm. This justifies the differentiation first on a dense smooth domain and then in the weak operator sense.

Set $\mathsf{C}_t = D_t R_t D_t^*$. The projection argument above applies at each time, so $(I - 2\mathsf{C}_t)$ is an isometry on the vector Hilbert space. Combining the last two derivatives gives the reflection identity

$$\frac{\mathrm{d}}{\mathrm{d}t}(D_t R_t) = c_t (I - 2\mathsf{C}_t) H_t D_t R_t, \qquad \left\| \frac{\mathrm{d}}{\mathrm{d}t}(D_t R_t) \right\| \leqslant |c_t|. \tag{5.4}$$

For the norm bound, $0 \preceq H_t \preceq I$ and (5.1) imply $\|H_t D_t R_t g\|_2 \leqslant \|H_t^{1/2} D_t R_t g\|_2 \leqslant \|g\|_2$. In particular the difference of the operators $D_t R_t$ at two times has norm at most the integral of $|c_t|$. The Gaussian map J_P is unchanged by the spatial pullback. Taking adjoints in $\mathcal{A}_t^* = J_P D_t R_t \Pi$ therefore gives

$$\|\dot{\mathcal{A}}_t\| \leqslant |c_t|, \qquad |\langle \dot{\mathcal{A}}_t f, f\rangle| \leqslant |c_t| \|\Pi f\|_2 \|(I - \Pi) f\|_2. \tag{5.5}$$

This calculation is the reason the moving corrector costs no inverse power of the spectral gap.

5.2 Reheating with a retained hidden Gaussian

Lemma 5.1 (Gaussian reheating)**.** *Let $\mu_u = \mu * \mathcal{N}(0, uI)$, $0 \leqslant u \leqslant t$, and assume that their potentials have Hessians between mI and I. Use the same $\varepsilon_0 = \theta\gamma$ in their correctors. From reference input $(X, P, E) \sim \mu \otimes \gamma_d \otimes \gamma_d$, send*

$$(X, P, E) \longmapsto (X + \sqrt{t}(nE + sG), P), \qquad G \sim \gamma_d \text{ independent.}$$

Let H_t^* be its density operator and put $v = \mathsf{H}_t^* f$, $D = \|f\|_2^2 - \|v\|_2^2$. If $\varepsilon_0\sqrt{t} \leqslant 1/24$, then

$$\mathcal{E}_{\mu_t}(v) \leqslant \mathcal{E}_\mu(f) - D/4 + C_H(\varepsilon_0 + \varepsilon_0^2)t\|f\|_2^2, \tag{5.6}$$

for an absolute C_H. Moreover $D \geqslant s^2\|(I - \Pi_E)f\|_2^2$, where Π_E averages only E.

Proof. We first calculate the effect of ordinary simultaneous heat on a relative density $g(x, P)$. If the reference density is p_u and the numerator is $(gp_0) * \mathcal{N}(0, uI)$, their quotient g_u satisfies

$$\partial_u g_u = \tfrac{1}{2}\Delta g_u - \nabla U_u \cdot \nabla g_u.$$

The reference continuity velocity is $\nabla U_u/2$. In its transported Hilbert space this becomes $\dot g_u = -L_u g_u/2$, while (5.4) applies with $c_u = -1/2$. Consequently

$$\frac{\mathrm{d}}{\mathrm{d}u}\|g_u\|_2^2 = -\|D_u g_u\|_2^2, \qquad D_g := \|g\|_2^2 - \|g_t\|_2^2 = \int_0^t \|D_u g_u\|_2^2 \,\mathrm{d}u.$$

Differentiating $J_u(g_u) = \langle \mathcal{A}_u g_u, g_u\rangle$ and integrating the two L_u terms by parts gives

$$\dot J_u(g_u) = \langle \dot{\mathcal{A}}_u g_u, g_u\rangle - \tfrac{1}{2}\langle D_u g_u, D_u \mathcal{A}_u^* g_u\rangle - \tfrac{1}{2}\langle D_u \mathcal{A}_u g_u, D_u g_u\rangle.$$

By equations (5.2) and (5.5), its absolute value is at most $\|g_u\|_2^2/4 + \|D_u g_u\|_2\|g_u\|_2$. Integration and Cauchy–Schwarz prove

$$|J_t(g_t) - J_0(g)| \leqslant (t/4)\|g\|_2^2 + \sqrt{tD_g}\|g\|_2. \tag{5.7}$$

Now retain the possible dependence of f on the old hidden coordinate E. Decompose $f = g + z$, where $g = \Pi_E f$. The channel in the lemma factors into the Gaussian OU operator O_n on E, with correlation coefficient n, followed by the map $(x, P, e) \mapsto (x + \sqrt{t}e, P)$. Set $w = \mathsf{O}_n z$. Gaussian Poincaré, or the Hermite expansion, gives $\|w\|_2 \leqslant n\|z\|_2$ and $\Pi_E w = 0$.

Let B pull output test functions back by this last map:

$$(B\phi)(x, P, e) = \phi(x + \sqrt{t}e, P).$$

It is an isometry from $L^2(\mu_t \otimes \gamma_d)$ into the input space, and $\mathsf{P}_B = BB^*$ is an orthogonal projection. The output relative density is $v = B^*(g + w)$. Thus the actual loss is exactly

$$D = (\|z\|_2^2 - \|w\|_2^2) + \|(I - \mathsf{P}_B)(g + w)\|_2^2 \geqslant s^2\|z\|_2^2 + \|(I - \mathsf{P}_B)(g + w)\|_2^2.$$

Also $v_0 = B^* g$ is ordinary simultaneous heat applied to g, so

$$D_g = \|(I - \mathsf{P}_B)g\|_2^2 \leqslant 2\|(I - \mathsf{P}_B)(g + w)\|_2^2 + 2\|w\|_2^2 \leqslant (2/s^2)D = 8D.$$

This proves the asserted lower bound on D and controls the ordinary-heat loss by the actual correlated-channel loss.

We next bound every additional corrector term. For any output test ϕ, Gaussian Poincaré in e gives

$$\|(I - \Pi_E)B\mathcal{A}_t\phi\|_2^2 \leqslant t\|D\mathcal{A}_t\phi\|_2^2 \leqslant t\|\phi\|_2^2.$$

The first equality of reference marginals here is $X + \sqrt{t}E \sim \mu_t$; the same argument gives $\|(I - \Pi_E)B\mathcal{A}_t^*\phi\|_2 \leqslant \sqrt{t}\|\phi\|_2$. Put $u = B^* w$, so $v = v_0 + u$. Since $\Pi_E w = 0$,

$$|\langle \mathcal{A}_t v_0, u\rangle| = |\langle (I - \Pi_E)B\mathcal{A}_t v_0, w\rangle| \leqslant \sqrt{t}\,\|v_0\|_2\|w\|_2,$$

$$|\langle \mathcal{A}_t u, v_0\rangle| = |\langle w, (I-\Pi_E)B\mathcal{A}_t^* v_0\rangle| \leqslant \sqrt{t}\,\|v_0\|_2\|w\|_2,$$
$$|\langle \mathcal{A}_t u, u\rangle| = |\langle (I-\Pi_E)B\mathcal{A}_t u, w\rangle| \leqslant \sqrt{t}\,\|u\|_2\|w\|_2 \leqslant \sqrt{t}\,\|w\|_2^2.$$

The input corrector averages E, hence $J_0(f) = J_0(g)$. Combining these three estimates with (5.7), $\|g\|_2 \leqslant \|f\|_2$, $D_g \leqslant 8D$, and $\|w\|_2^2 \leqslant (n^2/s^2)D = 3D$, gives

$$|J_t(v) - J_0(f)| \leqslant (t/4)\|f\|_2^2 + 7\sqrt{tD}\|f\|_2 + 3\sqrt{t}D.$$

Multiplying by ε_0, use

$$7\varepsilon_0\sqrt{tD}\|f\|_2 \leqslant D/8 + 98\varepsilon_0^2 t\|f\|_2^2, \qquad 3\varepsilon_0\sqrt{t}D \leqslant D/8.$$

Together with the exact half-norm loss $-D/2$, these inequalities prove (5.6), for example with $C_H = 100$. □

5.3 Transport, one collision, and random stopping

Proposition 5.2 (Contraction of the conditional lift)**.** *Before drawing the clock, split the entering momentum using a fresh $E_0 \sim \gamma_d$:*

$$Y_0 = sP + nE_0, \qquad H_0 = nP - sE_0, \qquad P = sY_0 + nH_0.$$

Draw $R \sim \mathrm{Unif}[0, \rho_*]$ independently. Starting from $X_0 = X$, evolve

$$X_\rho' = \ell Y_\rho - a\rho\nabla U_\rho(X_\rho), \qquad Y_\rho' = -\ell\nabla U_\rho(X_\rho). \tag{5.8}$$

For each fixed ρ, let Q_ρ be a conditionally Gaussian-invariant transition from H_0 to H_1, whose parameters may depend on the exact parent path up to ρ. Assume the resulting measurable kernels satisfy

$$\sup_{\rho \leqslant \rho_*} \|(Q_\rho - I)\mathcal{A}_\rho^*\| \leqslant C_Q a, \tag{5.9}$$

where C_Q is an absolute constant. At time R, set $P' = sY_R + nH_1$, $E' = -nY_R + sH_1$, reheat by

$$X' = X_R + \ell R(nE' + sG_x), \qquad G_x \sim \gamma_d,$$

and apply momentum OU for time Γ. For a sufficiently small absolute h, the resulting kernel P preserves $\mu_0 \otimes \gamma_d$ and

$$\mathcal{E}_{\mu_0}(\mathsf{P}^* f) \leqslant (1 + c\Gamma)^{-1}\mathcal{E}_{\mu_0}(f) \tag{5.10}$$

for every centered $f \in L^2(\mu_0 \otimes \gamma_d)$, with absolute $c > 0$. The small upper bound on h may depend on the declared absolute constant C_Q.

Proof. Reference laws and the common initial density. Since $\partial_\rho \mu_\rho = -a\rho\Delta\mu_\rho$, its continuity velocity is $-a\rho\nabla U_\rho$. The Hamiltonian part of (5.8) has zero divergence against $\mu_\rho \otimes \gamma_d$ at fixed ρ. Thus the full flow transports $\mu_0 \otimes \gamma_d$ to $\mu_\rho \otimes \gamma_d$. Uniqueness follows from its globally Lipschitz, linearly growing vector field on this compact clock interval.

The parent flow is invertible. Its past path is therefore determined by (X_ρ, Y_ρ), so Q_ρ is equivalently a kernel on H at fixed (X_ρ, Y_ρ). Conditional Gaussian invariance preserves the product reference $\mu_\rho \otimes \gamma_d \otimes \gamma_d$. The rotation to P', E' preserves the Gaussian product. Reheating adds variance $a\rho^2$, changing $\mu_\rho = \pi_{V_{a(1-\rho^2)}}$ back to $\mu_0 = \pi_{V_a}$, independently of P'. Momentum OU preserves this final reference. This proves invariance.

Use throughout the rotated coordinates

$$P = sY + nH, \qquad E = -nY + sH, \qquad Y = sP - nE.$$

At time zero the extended density is $f_0(x, Y, H) = f(x, sY + nH)$, independently of the clock. Its norm and modified energy are those of the supplied visible density. Let f_ρ be its density during the deterministic parent flow. Reference transport is unitary, so, with

$$M_\rho = \|\Pi f_\rho\|_2, \qquad R_\rho = \|(I - \Pi) f_\rho\|_2, \qquad N_0 = M_\rho^2 + R_\rho^2,$$

the number $N_0 = \|f\|_2^2$ is independent of ρ.

The transport current. At a fixed time, write

$$T_Y = Y \cdot \nabla_x - \nabla U_\rho \cdot \nabla_Y = sT_P - nT_E.$$

This operator is skew-adjoint under the instantaneous product law. For a centered scalar macro g, $J_P^*(T_Y g) = sDg$; hence $\mathcal{A}_\rho T_Y \Pi = s\Pi$. Also $\|T_Y \mathcal{A}_\rho r\|_2 = \|D\mathcal{A}_\rho r\|_2 \leqslant \|r\|_2$. For the remaining operator, put $u = L_\rho^{-1} g$. The adjoint of $\mathcal{A}_\rho T_Y (I - \Pi)$, applied to g, is

$$-(I - \Pi) T_Y (P \cdot \nabla u) = -s(PP^{\mathsf{T}} - I) : \nabla^2 u + nP^{\mathsf{T}} \nabla^2 u \, E.$$

The two Gaussian chaoses are orthogonal. For symmetric B, $\mathbb{E}(P^{\mathsf{T}} B P - \operatorname{tr} B)^2 = 2\|B\|_{\mathrm{HS}}^2$ and $\mathbb{E}(P^{\mathsf{T}} B E)^2 = \|B\|_{\mathrm{HS}}^2$. Thus its squared norm is $(2s^2 + n^2)\|\nabla^2 u\|_{\mathrm{HS},2}^2 \leqslant (1 + s^2)\|g\|_2^2$, by (5.1).

In the pure reverse-heat frame, the density derivative is $-\ell T_Y f_\rho$. Differentiating the corrector quadratic form, the Hamiltonian contribution is

$$\ell\{-\langle \mathcal{A}_\rho T_Y f_\rho, f_\rho\rangle + \langle T_Y \mathcal{A}_\rho f_\rho, f_\rho\rangle\} \leqslant \ell\{-sM_\rho^2 + \sqrt{1 + s^2} M_\rho R_\rho + R_\rho^2\} \leqslant \ell(-M_\rho^2/4 + 9R_\rho^2/4).$$

In the last step we used $s = 1/2$ and Young's inequality. The moving corrector contributes at most $a\rho M_\rho R_\rho$, by (5.5). If $\ell \leqslant 1/2$, then $a\rho \leqslant \ell/4$. Multiplying by ε_0, and using the constancy of N_0, therefore proves

$$\frac{\mathrm{d}}{\mathrm{d}\rho} \mathcal{E}_{\mu_\rho}(f_\rho) \leqslant -\frac{\varepsilon_0 \ell}{8} N_0 + 3\varepsilon_0 \ell R_\rho^2. \tag{5.11}$$

One collision estimate. The following calculation is for a fixed stopping time ρ. Momentum OU acts on P, whereas reheating uses only x, E. These channels commute, including as density operators between the two reference measures. We may consequently apply OU first. Set

$$q = Q_\rho^* f_\rho, \qquad o = \mathsf{O}_\Gamma q, \qquad v_\rho = \mathsf{H}_{a\rho^2}^* o,$$

and let D_Q, D_O, D_H be their respective squared-norm losses.

The dormant kernel leaves the position fixed and is conditionally Gaussian-invariant, so $\Pi q = \Pi f_\rho =: g_0$ and

$$R_\rho^2 = \|(I - \Pi) q\|_2^2 + D_Q.$$

Moreover, using the adjoint definition and $\Pi(Q_\rho - I)\mathcal{A}_\rho^* = 0$, we obtain

$$\begin{aligned} J_{\mu_\rho}(q) - J_{\mu_\rho}(f_\rho) &= \langle f_\rho, (Q_\rho - I)\mathcal{A}_\rho^* g_0\rangle \\ &= \langle (I - \Pi) f_\rho, (Q_\rho - I)\mathcal{A}_\rho^* g_0\rangle \leqslant C_Q a M_\rho R_\rho. \end{aligned}$$

It follows that

$$\mathcal{E}_{\mu_\rho}(q) \leqslant \mathcal{E}_{\mu_\rho}(f_\rho) - D_Q/2 + (C_Q/2)\varepsilon_0 a N_0.$$

To see exactly which microscopic directions the other two channels control, decompose $q = g + z$, $g = \Pi_E q$, and put $g_1 = g - \Pi_P g$. Since O_Γ commutes with Π_E,

$$D_O = (\|g\|_2^2 - \|\mathsf{O}_\Gamma g\|_2^2) + (\|z\|_2^2 - \|\mathsf{O}_\Gamma z\|_2^2).$$

The first summand is at least $(1 - e^{-2\Gamma})\|g_1\|_2^2$. The reheating lemma gives $D_H \geqslant s^2\|\mathsf{O}_\Gamma z\|_2^2$. For $0 < \Gamma \leqslant 1/2$, $1 - e^{-2\Gamma} \geqslant \Gamma$, so

$$\begin{aligned} D_O/2 + D_H/4 &\geqslant (\Gamma/2)\|g_1\|_2^2 + \tfrac{1}{2}(\|z\|_2^2 - \|\mathsf{O}_\Gamma z\|_2^2) + (s^2/4)\|\mathsf{O}_\Gamma z\|_2^2 \\ &\geqslant (\Gamma/2)\|g_1\|_2^2 + \tfrac{1}{16}\|z\|_2^2 \geqslant (\Gamma/16)\|(I - \Pi)q\|_2^2. \end{aligned}$$

Thus hidden momentum is paid by actual reheating loss, rather than being treated as independent under a nonstationary input law.

The corrector only uses the first P-Hermite component of its argument and the fully averaged macro. Therefore

$$J_{\mu_\rho}(\mathsf{O}_\Gamma q) = e^{-\Gamma} J_{\mu_\rho}(q).$$

Write $r = \|(I - \Pi)q\|_2$, $M = \|\Pi q\|_2$. The possible increase in the corrected part under OU is at most

$$\varepsilon_0(1 - e^{-\Gamma})|J_{\mu_\rho}(q)| \leqslant \theta\Gamma M r \leqslant (\Gamma/32)r^2 + 8\theta^2\Gamma M^2.$$

Finally, the reheating error in (5.6) is at most $C\theta\gamma a N_0$, since $a\rho^2 \leqslant a/4$, $\varepsilon_0 \leqslant \theta \leqslant 1/4$, and all these operators contract the uncorrected norm. The condition $\varepsilon_0\sqrt{a\rho^2} \leqslant 1/24$ holds after the absolute choices below. Combining the three channels, and using $D_Q/2 \geqslant (\Gamma/32)D_Q$, gives

$$\mathcal{E}_{\mu_0}(v_\rho) \leqslant \mathcal{E}_{\mu_\rho}(f_\rho) - (\Gamma/32)R_\rho^2 + (8\theta^2\Gamma + C_2\theta\gamma a)N_0, \tag{5.12}$$

where C_2 is absolute once C_Q is fixed.

Averaging the clock and choosing the constants. All v_ρ now have the common reference $\mu_0 \otimes \gamma_d$. Let $\overline{R^2} = \rho_*^{-1}\int_0^{\rho_*} R_\rho^2\,\mathrm{d}\rho$. Integrating (5.11) and then averaging the endpoint gives the exact survival-weight identity and its bound

$$\begin{aligned} \frac{1}{\rho_*}\int_0^{\rho_*} \mathcal{E}_{\mu_\rho}(f_\rho)\,\mathrm{d}\rho &\leqslant \mathcal{E}_{\mu_0}(f) - \frac{\varepsilon_0\ell\rho_*}{16}N_0 + \frac{3\varepsilon_0\ell}{\rho_*}\int_0^{\rho_*}(\rho_* - \rho)R_\rho^2\,\mathrm{d}\rho \\ &\leqslant \mathcal{E}_{\mu_0}(f) - c_0\theta\Gamma N_0 + 3\sqrt{2}\,\theta\Gamma\rho_*\overline{R^2}, \qquad c_0 = \sqrt{2}\,\rho_*/16. \end{aligned}$$

Average (5.12) and use convexity of the final energy, with $\mathsf{P}^* f = \rho_*^{-1}\int_0^{\rho_*} v_\rho\,\mathrm{d}\rho$. We obtain

$$\mathcal{E}_{\mu_0}(\mathsf{P}^* f) \leqslant \mathcal{E}_{\mu_0}(f) + \Gamma(3\sqrt{2}\,\theta\rho_* - 1/32)\overline{R^2} - \theta\Gamma(c_0 - 8\theta - 2C_2 h)N_0.$$

Choose once and for all

$$0 < \theta \leqslant \min\{1/4,\ (192\sqrt{2}\,\rho_*)^{-1},\ c_0/32\}, \qquad 0 < h \leqslant \min\{1/4,\ c_0/(8C_2)\}.$$

These choices imply $\ell \leqslant 1/2$, $\Gamma \leqslant 1/2$, and the reheating smallness condition. The coefficient of $\overline{R^2}$ is nonpositive, and $c_0 - 8\theta - 2C_2 h \geqslant c_0/2$. Consequently

$$\mathcal{E}_{\mu_0}(\mathsf{P}^* f) \leqslant \mathcal{E}_{\mu_0}(f) - (c_0\theta/2)\Gamma\|f\|_2^2.$$

Using (5.3) proves (5.10), after decreasing an absolute c. □

5.4 The dormant curvature certificate

We finish by deriving (5.9) from the actual conditional second-moment bound supplied by the sampling root. Fix the clock R and write $H_\rho(x) = \nabla^2 U_\rho(x)$. The root supplies deterministic quadrature nodes $\rho_i \leqslant R$ and positive weights with $\sum_i \omega_i = R$, and satisfies

$$\mathbb{E}[(n(H_1 - H_0))(n(H_1 - H_0))^{\mathsf{T}} \mid \text{exact parent path}] \preceq Ca^2 \sum_i \omega_i^2 H_{\rho_i}(X_{\rho_i}).$$

The harmless factors $1 - \rho_i^2$ in equation (12.22) have been dropped. This bound is evaluated only at exact reference centers.

Let $g_R \in L^2_0(\mu_R)$. Transport it backward by the pure reverse-heat flow $-a\rho\nabla U_\rho$ to obtain g_ρ; equivalently, its pullback to the fixed Hilbert space is constant. Thus $\|g_\rho\|_{L^2(\mu_\rho)} = \|g_R\|_2$. Put $u_\rho = L_\rho^{-1} g_\rho$ and $v_\rho = \nabla u_\rho$. The identity (5.4), returned to physical coordinates, reads

$$(\partial_\rho - a\rho\nabla U_\rho \cdot \nabla) v_\rho = a\rho(I - 2\mathsf{C}_\rho) H_\rho v_\rho, \qquad \mathsf{C}_\rho = \nabla L_\rho^{-1} \nabla^*.$$

Along the actual parent path there is additionally its Hamiltonian position velocity, so

$$\frac{\mathrm{d}}{\mathrm{d}\rho} v_\rho(X_\rho) = \ell \nabla^2 u_\rho(X_\rho) Y_\rho + a\rho[(I - 2\mathsf{C}_\rho) H_\rho v_\rho](X_\rho).$$

Since $(X_\rho, Y_\rho) \sim \mu_\rho \otimes \gamma_d$, Gaussian isometry and (5.1) bound the first term in RMS by $\ell\|g_R\|_2$. The reflection estimate bounds the second by $a\rho\|g_R\|_2$. For $0 \leqslant u \leqslant R$, integration gives

$$\|v_R(X_R) - v_u(X_u)\|_2 \leqslant \{\ell(R - u) + a(R^2 - u^2)/2\}\|g_R\|_2.$$

Using $H_u \preceq I$ and Bochner once more, we conclude

$$\|H_u(X_u)^{1/2} \nabla L_R^{-1} g_R(X_R)\|_2 \leqslant (1 + \ell + a)\|g_R\|_2. \tag{5.13}$$

This compares the final Poisson gradient with earlier curvature without comparing Hessians at unrelated points.

For a general extended test g, put $g_R = \Pi g - \int \Pi g \, \mathrm{d}\mu_R$ and $u_R = L_R^{-1} g_R$. Since $\mathcal{A}_R^* g = P \cdot \nabla u_R(X_R)$, the dormant transition changes this test by the conditional mean of $n(H_1 - H_0) \cdot \nabla u_R(X_R)$. Conditional Jensen followed by the displayed second-moment bound therefore gives

$$\begin{aligned}
\|(Q_R - I)\mathcal{A}_R^* g\|_2^2 &\leqslant \mathbb{E}\Big[\nabla u_R(X_R)^{\mathsf{T}} \mathbb{E}[(n\Delta H)(n\Delta H)^{\mathsf{T}} \mid \text{exact parent path}] \nabla u_R(X_R)\Big] \\
&\leqslant Ca^2 \sum_i \omega_i^2 \|H_{\rho_i}(X_{\rho_i})^{1/2} \nabla u_R(X_R)\|_2^2 \\
&\leqslant Ca^2 \Big(\sum_i \omega_i^2\Big)(1 + \ell + a)^2 \|\Pi g\|_2^2 \leqslant C' a^2 \|g\|_2^2.
\end{aligned}$$

This proves (5.9) uniformly in the clock. No reversibility or linear conditional mean of Q_R was used.

All density calculations above may first be made for smooth centered functions. Every potential in the clock interval is Gaussian-smoothed at a variance at least $3a/4 > 0$; its Hessian bounds ensure the stated complete flows, Poincaré inequalities, and closed Poisson inverses. Spatial integrations by parts can be justified with cutoffs and then the closed Dirichlet-form approximation used in (5.1). The resulting estimates are bounds between L^2 operators: the reference transports are unitary, the Markov operators are contractions, and the correctors are bounded. Density approximation therefore extends the final inequalities to every centered L^2 input under the original C^2 assumptions.

6 The conditional-mean interface

All finite routines use one convex potential F with $0 \preceq \nabla^2 F \preceq \lambda I$, where $\lambda = \delta$ and $\delta = e^{-y}$. Only ∇F is evaluated. For $0 < r \leqslant 1$, set

$$\nu_{x,r}(dz) \propto e^{-|z|^2/2 - F(x+rz)} dz, \quad g = \nabla F, \quad u_{x,r} = \mathbb{E}_{\nu_{x,r}} g(x + rZ), \quad D_F(x) = \sqrt{d} + |g(x)|. \tag{6.1}$$

Thus $x + rZ \sim R^F_{r^2,x}$ and $u_{x,r} = \nabla F_{r^2}(x)$. We write $L = \lambda r$ and $l = \lambda r^2$ and always take $l \leqslant 1/2$. The center x is arbitrary. Scalar constants and computation shapes are uniform in F, x, r, and are fixed before the oracle answers. An L^2 norm below is over the displayed exact reference law; for tensors we specify the operator or Hilbert–Schmidt norm.

Lemma 6.1 (Primitive moment bounds)**.** *For $Z \sim \nu_{x,r}$ and $G \sim \gamma_d$,*

$$\|Z\|_2 \leqslant \sqrt{d} + r|g(x)| \leqslant D_F(x), \qquad W_2(\nu_{x,r}, \gamma_d) \leqslant 2rD_F(x), \qquad \|g(x + rG) - u_{x,r}\|_2 \leqslant 3LD_F(x).$$

Moreover $u_{x,r}$ is λ-Lipschitz.

Proof. The density potential of $\nu_{x,r}$ has curvature at least one. Its mode z_* obeys $|z_*| \leqslant r|g(x)|$, and integration by parts gives $\mathbb{E}|Z - z_*|^2 \leqslant d$. Synchronous coupling of its Langevin diffusion with the Ornstein–Uhlenbeck diffusion gives $W_2(\nu_{x,r}, \gamma_d) \leqslant r\|g(x + rG)\|_2 \leqslant 2rD_F(x)$. Equivalently, differentiate the coupled squared distance, use strong monotonicity, and pass to stationarity. The third bound follows from the first and the L-Lipschitzness of $z \mapsto g(x + rz)$. The posterior Hessian identity gives $0 \preceq \nabla u \preceq \lambda I$. These elementary curvature consequences also follow from the Poincaré and synchronous-coupling toolkit in [Che26]. □

The construction has two public outputs. A mean call has a prescribed Gaussian input P and returns $\sigma P + \widehat{g}$; its reference is $u_{x,r} + \sigma P'$ for a Gaussian-preserving transition $P \mapsto P'$. A noisy sampling call $\mathsf{S}_\tau(x, r)$ approximates $\nu_{x,r} * \mathcal{N}(0, \tau^2 I)$, with a fresh final Gaussian reserve. Sections 7–12 construct these outputs and prove their retained-input error bounds. The single schedule in Section 13 pays all orders, shifts, errors, and query counts.

7 Gaussian lifting of conditional means

The mean routine uses an explicit Gaussian input p. Write its output as

$$\mathsf{M}_{x,r,\sigma}(p; \zeta) = \sigma p + \widehat{g}_{x,r,\sigma}(p; \zeta), \qquad u_{x,r} = \mathbb{E}_{R^F_{r^2,x}} \nabla F, \qquad S_{x,r} = \mathrm{Cov}_{R^F_{r^2,x}} (\nabla F).$$

The exact reference for $\widehat{g}$ is a Gaussian-preserving momentum correction. Its natural error budget is the posterior variance $S_{x,r}$. Scalar routine labels are fixed throughout; the reference kernel may depend on them. The finite implementation and its quantitative accuracy are given in proposition 12.5.

Gaussian assembly. Suppose $s^2 + \sum_i \omega_i^2 \sigma_i^2 + \beta^2 = \sigma^2$ with $s = \sigma/2$ and $\beta > 0$. For $a = \sigma^{-1}(s, \omega_1\sigma_1, \ldots, \omega_m\sigma_m)$, split the prescribed input p by

$$(G_0, p_1, \ldots, p_m) = a \otimes p + (I - aa^{\mathsf{T}})^{1/2} Z, \qquad Z \sim \gamma_{(m+1)d}. \tag{7.1}$$

Under standard Gaussian input P, these active coordinates are independent standard Gaussians. The comparison coordinate

$$G_E = \beta^{-1} \left(\sigma P - sG_0 - \sum_i \omega_i \sigma_i p_i \right)$$

is standard Gaussian independent of them. To check this explicitly, write $A = (G_0, P_1, \dots, P_m)$ and suppress I_d in block covariance matrices. Joint Gaussianity and the split formula give $\mathrm{Cov}(A) = I$ and $\mathrm{Cov}(A, P) = a$. Since $\sigma^2(1 - |a|^2) = \beta^2$, one obtains

$$\mathrm{Cov}(G_E, A) = 0, \qquad \mathrm{Cov}(G_E) = I, \qquad \sigma P = sG_0 + \sum_i \omega_i \sigma_i P_i + \beta G_E.$$

This identifies the full joint law with independent Gaussian coordinates conditioned on their prescribed affine combination. For each fixed input p the program uses exactly (7.1); the comparison coordinate G_E is recovered from that same input and is not resampled. The construction is local to one mean call.

Lemma 7.1 (Gaussian lifting)**.** *Let S be a square-integrable random vector, let Y be an observation, and put $u = \mathbb{E}S$, $m(Y) = \mathbb{E}[S \mid Y]$, and $\Sigma(Y) = \mathrm{Cov}(S \mid Y)$. Suppose a stationary centering path has marginals $\mathsf{Law}(Y) \otimes \gamma_d$ and satisfies*

$$s(G_1 - G_0) = \int_0^1 (m(Y_t) - u)\, \mathrm{d}t.$$

At each deterministic quadrature time t_i, let the child kernel $Q_i(y; p, dp')$ depend on the parent path only through $y = Y_{t_i}$ and fixed scalar data. For every y it preserves γ_d and, under $P_i \sim \gamma_d$ and $P_i' \mid P_i \sim Q_i(y; P_i, \cdot)$, satisfies

$$\sigma_i^2 \mathbb{E}[(P_i' - P_i)(P_i' - P_i)^{\mathsf{T}}] \preceq \Sigma(y).$$

Let $\omega_i \geqslant 0$, $\sum_i \omega_i = 1$, and use the Gaussian assembly above. Draw the parent position independently of the split Gaussian inputs, apply the child kernels independently conditional on the parent path, and set

$$\sigma P' = sG_1 + \sum_i \omega_i \sigma_i P_i' + \beta G_E.$$

This defines a Gaussian-preserving kernel retaining its input P, with

$$\sigma^2 \mathbb{E}[(P' - P)(P' - P)^{\mathsf{T}}] \preceq \mathrm{Cov}(S). \tag{7.2}$$

Proof. We distinguish conditioning on the parent path from conditioning on the supplied input. First take $P \sim \gamma_d$, independent of the position anchor and of the auxiliary array in (7.1). The Gaussian assembly makes $(G_0, P_1, \dots, P_m, G_E)$ a product of standard Gaussian laws. The parent centering path depends only on its independent position anchor and G_0. Consequently, for the sigma-field

$$\mathcal{F} = \sigma\{(Y_t, G_t) : 0 \leqslant t \leqslant 1\},$$

the variables $P_1, \dots, P_m, G_E$ are still independent standard Gaussians and are independent of $\mathcal{F}$. Any additional randomness defining the parent path is also taken independent of these variables. Given $\mathcal{F}$, apply each child kernel to its own P_i, using independent child randomizations. It follows that the pairs (P_i, P_i') are conditionally independent, and each P_i' has conditional law γ_d. These assertions condition on the parent path, not on P.

The conditional product law of $(P_1', \dots, P_m', G_E)$ is constant as a function of the parent path. Thus this array is independent of G_1, whose marginal is standard Gaussian by parent stationarity. The variance identity in the assembly gives

$$\mathrm{Cov}\left(sG_1 + \sum_i \omega_i \sigma_i P_i' + \beta G_E\right) = \left(s^2 + \sum_i \omega_i^2 \sigma_i^2 + \beta^2\right) I = \sigma^2 I.$$

The summands are independent Gaussian vectors, so $P' \sim \gamma_d$. Disintegrating this construction with respect to P gives a Markov kernel $Q(P, dP')$ preserving γ_d. It retains the very same input P occurring in the affine identity of the assembly. Gaussian preservation means $\int \gamma_d(dp)Q(p,\cdot) = \gamma_d$; it does not assert that P' is Gaussian conditionally on each fixed p.

Set $D_0 = s(G_1 - G_0)$ and $D_i = \sigma_i(P_i' - P_i)$. For every i, both conditional marginal means vanish, hence $\mathbb{E}[D_i \mid \mathcal{F}] = 0$. Since D_0 is $\mathcal{F}$-measurable,

$$\mathbb{E}[D_0 D_i^{\mathsf{T}}] = 0, \qquad \mathbb{E}[D_i D_j^{\mathsf{T}} \mid \mathcal{F}] = 0 \quad (i \neq j).$$

The latter identity follows from conditional independence, with no independence required between P_i and P_i'. Subtracting the two affine assembly identities gives $\sigma(P' - P) = D_0 + \sum_i \omega_i D_i$. Therefore

$$\sigma^2 \mathbb{E}[(P' - P)(P' - P)^{\mathsf{T}}] = \mathbb{E}[D_0 D_0^{\mathsf{T}}] + \sum_i \omega_i^2 \mathbb{E}[D_i D_i^{\mathsf{T}}]. \tag{7.3}$$

The child variance hypothesis, applied with its path-fixed center, gives $\mathbb{E}[D_i D_i^{\mathsf{T}} \mid \mathcal{F}] \preceq \Sigma(Y_{t_i})$. For any deterministic vector v, the centering identity and scalar Jensen give

$$v^{\mathsf{T}} \mathbb{E}[D_0 D_0^{\mathsf{T}}] v = \mathbb{E}\left[\int_0^1 v \cdot (m(Y_t) - u)\, dt\right]^2 \leqslant \int_0^1 \mathbb{E}[v \cdot (m(Y_t) - u)]^2\, dt = v^{\mathsf{T}} \operatorname{Cov}(m(Y)) v.$$

Every Y_{t_i} has law $\mathsf{Law}(Y)$ by stationarity. Inserting these bounds into (7.3), and using the nonnegative weights with $\sum_i \omega_i^2 \leqslant 1$, yields

$$\begin{aligned}\sigma^2 \mathbb{E}[(P' - P)(P' - P)^{\mathsf{T}}] &\preceq \operatorname{Cov}(m(Y)) + \sum_i \omega_i^2 \mathbb{E}\Sigma(Y) \\ &\preceq \operatorname{Cov}(\mathbb{E}[S \mid Y]) + \mathbb{E} \operatorname{Cov}(S \mid Y) = \operatorname{Cov}(S).\end{aligned}$$

This is an exact-reference assertion. It uses neither the distribution of an implemented center nor an estimate of its numerical error. □

The posterior reference. Apply the lemma to $S = \nabla F(A)$ for $A = x + rZ \sim R^F_{r^2,x}$ and $Y = b_* Z + RN$, where $b_*^2 + R^2 = 1$ and N is independent Gaussian. Gaussian conditioning gives

$$\mathsf{Law}(A \mid Y = y) = R^F_{r^2 R^2, x + r b_* y}.$$

The analytic centering flow supplies the required stationary path. Take the identity Gaussian kernel at terminal mean calls and apply lemma 7.1 recursively to the final positive quadrature weights. It follows at every finite routine height that

$$\sigma^2 \mathbb{E}[(P' - P)(P' - P)^{\mathsf{T}}] \preceq S_{x,r}. \tag{7.4}$$

This reference uses only the final child kernels. Private center computations, numerical layers, and their errors do not enter its definition. In particular, no curvature bound is propagated through the recursion.

Finite corrections and curvature. The implementation in proposition 12.5 has the same local Gaussian assembly. At a terminal call $\widehat{g} = g(x + rZ)$; at a nonterminal call it is the weighted sum of the final child corrections. Its exact reference is

$$G = u_{x,r} + \sigma(P' - P), \qquad \mathbb{E}G = u_{x,r}, \qquad \operatorname{Cov}(G) \preceq S_{x,r}.$$

For $r^2\lambda \leqslant 1/2$, the posterior Poincaré calculation in lemma 4.1 converts the covariance once:

$$S_{x,r} \preceq 2r^2\lambda\nabla^2 F_{r^2}(x).$$

This is the conditional-gradient covariance principle of [CCLZ26a, Lemma C.4]. The recursion itself uses only total covariance; finite approximation errors are handled separately.

Affine output rows. Conditionally independent centered child increments obey the elementary closure rule

$$\mathbb{E}\left[\left(\sum_i C_i\sigma_i\Delta P_i\right)\left(\sum_i C_i\sigma_i\Delta P_i\right)^{\mathsf{T}}\middle|\mathcal{A}\right] \preceq \sum_i C_i S_i C_i^{\mathsf{T}}$$

whenever $\sigma_i^2\mathbb{E}[\Delta P_i\Delta P_i^{\mathsf{T}} \mid \mathcal{A}] \preceq S_i$. The sampling root therefore has the direct form $p + \sum_i b_i\widehat{g}_i$ and inherits its posterior variance budget from (7.4). This is the only additional estimate needed for the Gaussian momentum correction in the mixing proof.

8 Gaussian localization and probability transport

We use Gaussian localization, Poincaré, and integration by parts to construct the exact comparison flows. A single second-moment adjoint calculus then supplies all numerical regularity estimates. The construction is adapted from [Ope26]; Gaussian conditional estimates of this kind also appear in [CCLZ26b, Appendix C.2].

Retain the convex primitive F, its gradient g, and $L = \lambda r$, $l = rL \leqslant 1/2$ from section 7. Write $f(z) = g(x + rz)$, $u = \mathbb{E}_\nu f$, and, for independent $Z \sim \nu = \nu_{x,r}$ and $N \sim \gamma_d$,

$$\upsilon_\rho = 1 - \rho^2, \quad \xi_\rho = \mathsf{Law}(\rho Z + \sqrt{\upsilon_\rho}N), \quad M_\rho(y) = \mathbb{E}[f(Z) \mid \rho Z + \sqrt{\upsilon_\rho}N = y].$$

Incorporate normalization into $\xi_\rho(dy) = e^{-H_\rho(y)}dy$. Completing the quadratic gives

$$\mathsf{Law}(Z \mid Y = y)(dz) \propto \exp\left(-\frac{|z - \rho y|^2}{2\upsilon_\rho} - F(x + rz)\right)dz. \tag{8.1}$$

Thus M_ρ is the primitive mean at center $x + r\rho y$ and scale $r\sqrt{\upsilon_\rho}$, and $M_0 = u$. A tensor's *split norm* is its matrix operator norm after any nontrivial partition of its slots; the output slot is included. Tensor slots carry their Euclidean, or Hilbert–Schmidt, norm.

Lemma 8.1 (Poincaré controls all cumulant splits)**.** *Suppose $S \in \mathbb{R}^D$ has a moment-generating function near zero and satisfies $\operatorname{Var} P(S) \leqslant \beta\mathbb{E}|\nabla P(S)|^2$ for all polynomials. Then every nontrivial split of its cumulant tensor satisfies*

$$|\kappa_n(S)|_{\text{split}} \leqslant (n-1)!\beta^{n/2}, \qquad n \geqslant 2.$$

Proof. The Appell-polynomial device is classical [Ans04]; its use in the conditional cumulant argument follows [Ope26, Section 3]. Let P_j be the Appell tensors defined by the generating function $e^{\theta\cdot s}/\mathbb{E}e^{\theta\cdot S}$. For any coefficient tensor T, the lower derivatives of $T : P_j$ have mean zero, while $\nabla^j(T : P_j) = j!\operatorname{Sym} T$. Iterating Poincaré componentwise gives

$$\mathbb{E}|T : P_j(S)|^2 \leqslant \beta^j(j!)^2|T|_{\text{HS}}^2.$$

For a prescribed split $j \mid k$, the normalized mixed coefficients of

$$G(\theta,\omega)=\frac{\mathbb{E}e^{(\theta+\omega)\cdot S}}{\mathbb{E}e^{\theta\cdot S}\mathbb{E}e^{\omega\cdot S}}$$

therefore have split norm at most $\beta^{(j+k)/2}$, by Cauchy–Schwarz. Products of normalized coefficients are tensor products followed by separate symmetrization within the two groups; both operations preserve the required multiplicative norm bound. The mixed coefficient of $\log G$ is consequently majorized, after removing the power of β, by the corresponding coefficient of

$$-\log\left(1-\frac{st}{(1-s)(1-t)}\right)=\log(1-s)+\log(1-t)-\log(1-s-t).$$

That coefficient is $(j+k-1)!/(j!k!)$. Multiplying by $j!k!$ proves the claim. All coefficient calculations are finite; $\beta=0$ follows because the coordinate variances vanish. □

Proposition 8.2 (Gaussian conditional jets)**.** *For $q\geqslant 1$, $U_q=\rho^{-q}\nabla^q M_\rho$ extends smoothly to zero, and every split satisfies*

$$|U_q|_{\mathrm{split}}\leqslant 2^{q+1}q!L\upsilon_\rho^{-(q-1)/2},\qquad \|U_1\|_{\mathrm{op}}\leqslant 2L. \tag{8.2}$$

Moreover U_1 is symmetric, and

$$\nabla H_\rho=y+r\rho M_\rho,\qquad \|\nabla^2 H_\rho\|_{\mathrm{op}}\leqslant 2. \tag{8.3}$$

For $j\geqslant 1$, every split obeys

$$|\nabla^j\nabla^2 H_\rho|_{\mathrm{split}}\leqslant 2^{j+2}(j+1)!\upsilon_\rho^{-j/2}. \tag{8.4}$$

Proof. The conditional law in (8.1) has Poincaré constant at most υ_ρ. The map $z\mapsto(z,f(z)/L)$ is $\sqrt{2}$-Lipschitz, so its image satisfies Poincaré with constant $2\upsilon_\rho$. Linear-tilt differentiation gives

$$(U_q)_{i,k_1,\dots,k_q}=\upsilon_\rho^{-q}\operatorname{cum}(f_i(Z),Z_{k_1},\dots,Z_{k_q}\mid Y=y). \tag{8.5}$$

The cumulant lemma proves the factorial bound. Direct Poincaré and Cauchy–Schwarz give $\|U_1\|_{\mathrm{op}}\leqslant L$. Only the first derivative of f is used in pulling back Poincaré. Conditional integration by parts gives $\mathbb{E}[Z\mid Y=y]=\rho y-\upsilon_\rho r M_\rho(y)$, hence the score identity. Differentiating it proves symmetry and the remaining bounds. The conditional integrals and their parameter derivatives are smooth for $\rho<1$, including the cumulant extension at zero. Their derivatives have polynomial growth on compact probability-time intervals: the conditional mode has size $O(|y|+r|g(x)|)$ and the centered law has Gaussian tails. These facts justify all finite differentiations and cutoff integrations below under the original C^2 assumption on F. □

Define the row-wise adjoint divergence by $(\nabla^*V)_I=\sum_k[-\partial_k V_{I,k}+(\partial_k H_\rho)V_{I,k}]$; thus $\int(\nabla^*V)_I\phi\,d\xi_\rho=\int V_{I,k}\partial_k\phi\,d\xi_\rho$.

Proposition 8.3 (Probability transport)**.** *The complete Lipschitz flow*

$$Y_\rho'=-rM_\rho(Y_\rho) \tag{8.6}$$

transports ξ_a to ξ_b on compact intervals in $[0,1)$. Writing $\mathcal{D}=\partial_\rho-rM_\rho\cdot\nabla$, its conditional mean satisfies

$$\mathcal{D}M_\rho=\nabla^*U_1. \tag{8.7}$$

Proof. The interpolation's conditional velocity is $\mathbb{E}[(Z-\rho Y)/\upsilon_\rho \mid Y] = -rM_\rho(Y)$. Testing against a smooth compactly supported function gives its continuity equation. The velocity is $2l$-Lipschitz with linear growth, so uniqueness of transport by its complete flow identifies the marginals. For the f_i-weighted continuity equation, subtract the conditional velocity from the weighted flux. The remaining current is $\mathrm{Cov}(f_i(Z), Z \mid Y)/\upsilon_\rho = (U_1)_{i,\cdot}$. The adjoint identity gives (8.7). Equivalently, if $q_\rho = d\xi_\rho/d\gamma_d$, the velocity is $\rho^{-1}\nabla \log q_\rho$, the reverse Ornstein–Uhlenbeck flow. □

9 A transported Stein kernel and its Gaussian lift

The transported centering construction adapts [Ope26, Section 5]. For general Stein-kernel existence under a Poincaré inequality, see [CFP19]; that variational existence result is background, not the source of the specific transport formula below. Fix $0 < R < 1/2$, put $b_* = (1-R^2)^{1/2}$, and write a star for $\rho = b_*$. Let $T_\rho = T_{\rho\to b_*}$, $Y_\rho = T_\rho^{-1}$, and $B_\rho = DT_\rho \circ Y_\rho$. The inverse-flow equation gives

$$B_\rho' = r\rho B_\rho U_1(\rho, Y_\rho), \qquad B_{b_*} = I, \quad B_0' = 0, \quad \|B_\rho\|_{\mathrm{op}} \leqslant e. \tag{9.1}$$

Lemma 9.1 (A transported Stein kernel)**.** *The smooth field*

$$K^T = \int_0^{b_*} B_\rho U_1(\rho, Y_\rho)\, d\rho = \frac{1}{r}\int_0^{b_*} B_\rho' \frac{d\rho}{\rho} = \frac{1}{r}\int_0^{b_*} \log\frac{b_*}{\rho}\, B_\rho''\, d\rho \tag{9.2}$$

satisfies $\nabla_{\xi_*}^* K = M_* - u$ and $\|K\|_{\mathrm{op}} \leqslant 2eL$. The first integral defines the second integrand at zero.

Proof. For a compactly supported scalar test ϕ, the function $\phi \circ T_\rho$ has zero material derivative. The conditional-current identity therefore yields

$$\frac{d}{d\rho}\mathbb{E}_{\xi_\rho}[M_{\rho,i}\phi\circ T_\rho] = \mathbb{E}_{\xi_*}\sum_j (U_1 B_\rho^T)_{ij}(\rho, Y_\rho)\partial_j\phi.$$

Integrate from zero to b_*, using $M_0 = u$ and symmetry of U_1. This is the Stein identity. The norm bound follows from the first integral. Finally $B_0' = 0$ and Fubini give the logarithmic representation. □

Lemma 9.2 (Factorial spatial transport bounds)**.** *On any probability-transport interval with smoothing width at least $R \leqslant 1$, both orientations of transport satisfy, for $j \geqslant 1$,*

$$|\nabla^j T|_{\mathrm{split}} \leqslant C^{j+1} j! R^{1-j}, \qquad |\nabla^j (T - \mathrm{id})|_{\mathrm{split}} \leqslant l C^{j+1} j! R^{1-j}.$$

For the Stein kernel, $j \geqslant 0$ gives $|\nabla^j K|_{\mathrm{split}} \leqslant LC^{j+1} j! R^{-j}$. At $j = 0$ this is the matrix operator norm.

Proof. A single contraction retaining free slots in both factors is a composition of suitable split unfoldings, tensored with identities; its split norm is bounded by the product. Thus the differentiated flow equations, divided by $j!R^{1-j}$, are majorized coefficientwise by

$$\partial_t b = \frac{4lb}{1-2b}, \qquad b(0,z) = z.$$

For each finite Taylor truncation, positivity and a first-hitting argument at $1/4$ give $b(t, 1/256) \leqslant e^4/256 < 1/4$ for $0 \leqslant t \leqslant 1$. Its coefficients are bounded by 256^j; integrating the equation gives the factor l for $T - \mathrm{id}$. Reverse transport uses the same absolute interval lengths. For the kernel, take $a = 256$ and the positive majorants $P(z) = az/(1-az)$ and $Q(z) = 4/(1-2z)^2$ for transport and U_1/L. The normalized kernel derivatives are majorized by $P'(P(z))Q(P(z))$. Evaluation at $z = 1/(4a^2)$ bounds its degree-j coefficient by $81a(4a^2)^j$. These finite coefficient comparisons prove every split bound without any dimension factor. □

Proposition 9.3 (The Gaussian lift of the centered mean)**.** *For $s > 0$, the complete flow*

$$\dot{Y} = -K(Y)^T G/s, \qquad \dot{G} = (M_*(Y) - u)/s \tag{9.3}$$

preserves $\xi_* \otimes \gamma_d$. From an invariant starting point,

$$sG_0 + \int_0^1 M_*(Y_t)\, dt = u + sG_1 \overset{\text{law}}{=} u + sG, \qquad G \sim \gamma_d. \tag{9.4}$$

Proof. The smooth field has linear growth since K is bounded and M_* is Lipschitz, so Gronwall gives completeness in both directions. For $p(y,g) \propto e^{-H_*(y)-|g|^2/2}$, the Stein identity gives $\operatorname{div}_y(p\dot{Y}) = p\, g\cdot(M_* - u)/s = -\operatorname{div}_g(p\dot{G})$. The flow's Jacobian change of variables proves stationarity; integration of its momentum equation proves the output identity. □

Its generator is the skew-adjoint lift

$$\mathcal{L} = \nabla^* A \nabla, \qquad A = \begin{pmatrix} 0 & K^T/s \\ -K/s & 0 \end{pmatrix}. \tag{9.5}$$

Skew symmetry cancels the second-order part. Poincaré gives $\mathbb{E}|M_* - u|^2 \leqslant 2L^2 d$, and conditional Gaussian isometry gives $\mathbb{E}|K(Y_t)^T G_t|^2 = \mathbb{E}\|K(Y_t)\|_{\mathrm{HS}}^2 \leqslant 4e^2 L^2 d$. Consequently

$$\|(Y_t, G_t) - (Y_0, G_0)\|_2 \leqslant Ct(L/s)\sqrt{d}, \qquad 0 \leqslant t \leqslant 1. \tag{9.6}$$

10 One second-moment adjoint calculus

The adjoint-divergence and material-derivative framework adapts [Ope26, Section 4]; the present organization uses a common second-moment calculus for the numerical estimates. Only second moments on exact reference trajectories enter the numerical formulas. This section proves their dimension dependence directly. In particular, the tensor argument below does not invoke a bound at a moment order depending on the dimension. We use a common envelope

$$\mathcal{A}_m = [C(m+1)]^{C(m+1)}, \qquad m \geqslant 0,$$

where the absolute constant C is chosen once large enough for all the conclusions of this section. Replacing m by $cm + c$ for an absolute c, or multiplying finitely many such envelopes whose indices sum to $O(m)$, is absorbed by this choice of C.

10.1 Expressions, coordinate contractions, and their size

Fix a smooth probability law $\mu(dz) \propto e^{-H(z)}dz$ on $\mathbb{R}^D$ with Gaussian tails, and $0 < R \leqslant 1$. All fields and all derivatives used below have polynomial growth. Every slot of a tensor takes values in $\mathbb{R}^D$; tensors on coordinate subspaces can be extended by zero. For a nonempty proper set of slots I, write $|T|_{I|I^c}$ for the operator norm of the corresponding matrix unfolding. A bound on $|T|_{\mathrm{split}}$ means the bound for every such I, with the same constant.

A coefficient T is assigned an integer size $a \geqslant 2$, at least its number of slots, and a nonnegative weight w. It has at least two slots and satisfies, for every $j \geqslant 0$ and every nontrivial split,

$$|\nabla^j T|_{\mathrm{split}} \leqslant c^{a+j}(a+j)! R^{-w-j}. \tag{10.1}$$

Here $c \geqslant 1$ is absolute and common to all coefficients. We require $S = \nabla^2 H$ to be an available coefficient of size two and weight zero. In particular, the score has at most linear growth.

An expression is a finite formula made from coefficients by the following operations. Slot permutations do not change its size or weight. A contraction joins one slot of each of two expressions, leaving at least one free slot in each operand; its weight is the sum of the weights. A spatial derivative adds a labeled slot and increases weight by one. An adjoint divergence removes one designated slot, leaves at least one free slot, and increases weight by one. The divergence convention is

$$(\nabla^* V)_I = \sum_{k=1}^{D} [-\partial_k V_{I,k} + H_k V_{I,k}].$$

The size $n(E)$ is the sum of all coefficient sizes and the number of contraction, derivative, and adjoint nodes. Repeated occurrences of a formula count separately. Thus the formula is a tree, even if two of its coefficient values happen to agree. Scalar expressions and traces within one operand are not operations of this grammar.

We first record the coordinate ordering used in the proof. Consider a complete coordinate contraction of tensor fields, cut one coordinate edge, and order the tensor occurrences. Regard the cut coordinate as an external input at the first occurrence and an external output at the last. Every other edge is directed from its earlier endpoint to its later endpoint. Call the order admissible if every occurrence has at least one incoming and at least one outgoing slot.

Lemma 10.1 (An ordered contraction has one dimension factor)**.** *For a contraction with an admissible order, its absolute value at each point is at most*

$$D \prod_v |T_v|_{\mathrm{split}}.$$

An occurrence in such an order may be replaced by two consecutively ordered occurrences joined on one new coordinate, provided both inherit at least one old incident slot. The order can be chosen admissibly. Adding an edge between two distinct occurrences also preserves admissibility when directed according to their existing order.

Proof. For the replacement assertion, call the two children A and B. If A has only old incoming slots, place it first: the new joining edge supplies an outgoing slot. If it has only old outgoing slots, place it second. Impose the same requirement for B. The requirements cannot conflict. Two children both having only incoming slots would leave the old parent without an outgoing slot, and the analogous statement holds for outgoing slots. A child with both kinds imposes no requirement. A child lacking one kind receives that kind from the joining edge. All other occurrences keep all their old slots, so their conditions are unchanged. Adding an edge removes no old slot and creates no problem.

To prove the bound, process the occurrences in their order. Immediately before each occurrence retain the indices of all edges that have entered the computation but have not yet been consumed. Initially there is only the cut input, so the initial space is $\mathbb{R}^D$. Permute the retained tensor coordinates, apply the next tensor's unfolding to its incoming indices, and apply the identity to all other retained indices. Its outgoing indices enter the new retained list. Both groups for the tensor are nonempty, so its norm is one of the split norms in the hypothesis. Coordinate permutations and tensor products with identities have norm one. Matrix multiplication performs exactly the sums on consumed indices. At the end only the cut output remains. The composition is an operator $Q : \mathbb{R}^D \to \mathbb{R}^D$ with $\|Q\| \leqslant \prod_v |T_v|_{\mathrm{split}}$. Restoring the cut coordinate is $\operatorname{tr} Q$, whose absolute value is at most $D\|Q\|$. Intermediate spaces can have more than one tensor coordinate; their dimensions do not enter this operator-norm calculation. □

Lemma 10.2 (Second-moment adjoint calculus)**.** *For every expression E and every $j \geqslant 0$,*

$$\|\nabla^j E\|_{\mathrm{HS},2} \leqslant \sqrt{D}\, \mathcal{A}_{n(E)+j} R^{-w(E)-j}. \tag{10.2}$$

For vector output the Hilbert–Schmidt norm is the Euclidean norm. The constant is independent of the formula, dimension, and derivative order, apart from the displayed size dependence.

Proof. Put $N = n(E) + j$. We give the coordinate expansion, its invariant, its number of terms, and its smoothing weight separately.

Initial contraction. Suppose first that $\nabla^j E$ has $r \geqslant 2$ free slots. Write its squared Hilbert–Schmidt norm as two labeled copies joined at corresponding slots, and regard each copy temporarily as an opaque occurrence. Cut one of the r joining coordinates. Order the first copy before the second, directing the remaining $r-1$ coordinates forward. The first copy receives the cut input; the second emits the cut output. This is an admissible order. Opaque here means that the expression defining a tensor has not yet been expanded; it is still an ordinary tensor field.

Reduction rules. Decorate each opaque occurrence with the ordered, individually labeled spatial derivatives applied to it. A derivative node is absorbed into this list. When the outer operation is a contraction, Leibniz's rule assigns each derivative label to one child, and replaces the occurrence by the two differentiated children and their joining coordinate. Each child inherits its original free slots, at least one by the grammar. The replacement assertion of lemma 10.1 therefore maintains the admissible order for every Leibniz term.

For an occurrence $\partial_J \nabla^* V$, the exact commutation formula is

$$\partial_J \nabla^* V = \nabla^*(\partial_J V) + \sum_{\varnothing \neq A \subseteq J} (\partial_A \nabla H) * \partial_{J \setminus A} V. \tag{10.3}$$

Subsets refer to labeled derivative positions, so this formula already includes repeated-index multiplicities. In a score term, the score factor inherits the nonempty old group A, and the operand inherits the original output slots of the adjoint, at least one. They are joined on the former divergence coordinate. This is another admissible two-child replacement. The new score tensor is a derivative of $S = \nabla^2 H$, because $|A| \geqslant 1$.

For the principal term in (10.3), remove the outer adjoint by integration by parts in the complete integral. For any product Ψ of the other occurrences the identity is

$$\int (\nabla^* \partial_J V)_I \Psi \, d\mu = \sum_k \int (\partial_J V)_{I,k} \partial_k \Psi \, d\mu.$$

Leibniz's rule selects one of the other opaque occurrences for the new derivative. In particular it never differentiates its own operand. The former divergence slot and the new derivative slot form an edge between distinct occurrences. Both retain all their old slots, so orienting this edge in the existing order preserves admissibility. Only after this operation do we expand the operand V.

Apply these rules in a fixed deterministic order until all occurrences are derivatives of coefficients or of S. They terminate: a derivative label moves down a finite formula, every contraction is expanded once, and every adjoint is removed once, by either integration by parts or a score term. No rule creates a new adjoint. The admissible order has been preserved at every step, so each terminal contraction costs exactly the single dimension factor of lemma 10.1.

Counting without repeated Leibniz losses. There are at most $2N$ original operation nodes across the two copies, and their coefficient sizes sum to at most $2N$. The initial external derivatives and the derivative nodes contribute at most $2N$ labels. Each adjoint removed by integration by parts creates one additional label; there are at most $2N$ such adjoints. Hence at most $4N$ derivative labels

occur. A label is routed, never copied, within any single term. At most $2N$ score occurrences are created, each attached to the address of the unique original adjoint that created it.

Here is an explicit encoding of a terminal summand, including its multiplicity. Give every original formula node its tree address and every derivative its origin label. For each original adjoint record whether it was removed by integration by parts or by a score term. For each integration-by-parts removal record its selected recipient occurrence, using its original tree address or its creating-adjoint address if it is a score occurrence. For each derivative label record its final destination, an original coefficient leaf or one of the score occurrences. There are $O(N)$ possible addresses of each kind.

This record reconstructs a labeled expansion path under the fixed reduction order. At a contraction, the destination determines the child receiving each label. At a score commutator, precisely the labels destined for that score occurrence comprise the nonempty subset A; the others stay with the operand. At an integration-by-parts step its recorded recipient determines where the newly born label starts. The subsequent routing is again determined by its destination. Thus different labeled expansion paths have different records. The total number of records is bounded by

$$2^{2N}(CN)^{2N}(CN)^{4N} \leqslant [C(N+1)]^{C(N+1)}.$$

This counts all Leibniz and commutator multiplicities; one does not multiply an additional binomial bound each time a label passes through an internal node.

Coefficient sizes and smoothing weights. The original coefficient sizes sum to at most $2N$. Every new score occurrence is $\nabla^h S$; its base size is two, and its derivatives are among the labels already counted (one derivative of the score is used to identify S). Consequently the sum of terminal coefficient sizes and derivative orders is at most CN. If their individual sizes plus orders are b_v, then

$$\prod_v c^{b_v} b_v! \leqslant c^{\sum_v b_v} \Big(\sum_v b_v\Big)! \leqslant [C(N+1)]^{C(N+1)}.$$

For the powers of R, initially the two copies have total weight $2(w(E)+j)$. A contraction or spatial-derivative routing preserves this weight. Integration by parts deletes one adjoint of weight one and adds one derivative elsewhere. A score term with $a = |A| \geqslant 1$ replaces weight $w(V)+1+|J|$ by

$$(a-1) + w(V) + |J| - a = w(V) + |J| - 1.$$

It decreases the weight by two. Therefore the terminal power is at most $2(w(E)+j)$. Since $R \leqslant 1$, the ordered-contraction bound, the coefficient bound, and the term count give

$$\int |\nabla^j E|_{\mathrm{HS}}^2 \, d\mu \leqslant D[C(N+1)]^{C(N+1)} R^{-2(w(E)+j)}.$$

Taking the square root proves the desired estimate when $r \geqslant 2$.

Vector output. The remaining case is $j = 0$ and E has one free slot. After ignoring slot permutations its outermost operation must be an adjoint: coefficients have at least two slots, a derivative adds a slot, and a permitted contraction retains at least two slots in total. Write $E = \nabla^* V$, where V has two slots. Start with

$$\int |E|^2 \, d\mu = \int \sum_{i,k} V_{i,k} \partial_k (\nabla^* V)_i \, d\mu.$$

The two opaque occurrences V and $\nabla(\nabla^* V)$ are joined on the coordinates i, k. Cutting either one gives an admissible initial order. Now use exactly the reductions above. The first integration by parts has consumed one of the original adjoints and created one of the counted derivative labels, so

both the size bound and the weight bound are unchanged. In particular, in a subsequent score split its derivative slot and the remaining output slot give each child an old slot. This proves the vector estimate as well.

Finally, every identity can first be performed with a compactly supported cutoff on the full coordinate integrand. Terms where a derivative hits the cutoff are polynomially bounded on its escaping support and vanish by Gaussian tails. There are only finitely many terms for each N, D. Letting the cutoff radius tend to infinity proves the stated integrations by parts; the uniform constants come from the finite bounds just established. □

10.2 Material and Jacobian derivatives

For ξ_ρ, take $R = \varsigma = \sqrt{1-\rho^2}$. The normalized jets U_q/L are coefficients of size $q+1$ and weight $q-1$: $\nabla^j U_q = \rho^j U_{q+j}$ and the factorial jet bound verify (10.1), after increasing the absolute c. The same is true of the score Hessian by (8.4).

Lemma 10.3 (Material identities)**.** *For labeled derivative slots I, $|I| = q \geqslant 1$, one has*

$$(\mathcal{D}U_q)_{i,I} = (\nabla^* U_{q+1})_{i,I} + 2r\rho \sum_{\varnothing \neq A \subseteq I} (U_{|A|})_{k,A} (U_{q-|A|+1})_{i,k,I\setminus A}, \tag{10.4}$$

$$[\mathcal{D}, \nabla^*]V = r\nabla^*((DM_\rho)V). \tag{10.5}$$

Here DM_ρ denotes the spatial Jacobian, and the contracted slot on the right of the second identity is the divergence slot of V.

Proof. Work first at $\rho > 0$. Since $\mathcal{D} = \partial_\rho - rM_k\partial_k$, Leibniz gives

$$\mathcal{D}\partial_I M_i = \partial_I(\mathcal{D}M_i) + r \sum_{\varnothing \neq A \subseteq I} (\partial_A M_k)\partial_{k,I\setminus A} M_i.$$

Substitute $\mathcal{D}M = \nabla^* U_1$ and commute ∂_I past this adjoint using (10.3). The principal term, after division by ρ^q, is $\nabla^* U_{q+1}$. For a singleton A, differentiating the linear part of $\nabla H_\rho = y + r\rho M_\rho$ gives $q\rho^{-1}U_q$ in total. It cancels the derivative of the factor ρ^{-q} in U_q. For every nonempty A, the nonlinear score term is

$$\rho^{-q} r\rho\, \rho^{|A|} U_{|A|} \rho^{q-|A|} U_{q-|A|+1} = r\rho U_{|A|} * U_{q-|A|+1}.$$

The transport commutator gives the identical term, since its second factor has $q-|A|+1$ derivatives of M. Adding them proves the first identity, including its factor two.

For the second identity, differentiate $\int (\nabla^* V)\phi\, d\xi_\rho = \int V_k \partial_k \phi\, d\xi_\rho$ along the probability transport. The continuity equation makes differentiation of an expectation equal to expectation of the material derivative. The terms containing $\mathcal{D}\phi$ cancel by the adjoint identity. The remaining commutator is $[\mathcal{D}, \partial_k]\phi = r(\partial_k M_j)\partial_j \phi$. It follows that

$$\int \big(\mathcal{D}\nabla^* V - \nabla^* \mathcal{D} V\big)\phi\, d\xi_\rho = r \int V_k (\partial_k M_j)\partial_j \phi\, d\xi_\rho,$$

which is the second formula. Smoothness and polynomial growth justify these identities against cutoffs. The nonsingular jet extensions give both identities at $\rho = 0$ by continuity. □

Corollary 10.4 (The derivative bounds used by quadrature)**.** *For $m \geqslant 1$,*

$$\|\mathcal{D}^m M_\rho\|_2 \leqslant L\sqrt{d}\, \mathcal{A}_m \varsigma^{-(2m-1)}. \tag{10.6}$$

For $y \sim \xi_*$ and an independent $G \sim \gamma_d$,

$$\|B_\rho^{(m)}\|_{\mathrm{HS},2} = \|(B_\rho - I)^{(m)} G\|_2 \leqslant l\sqrt{d}\, \mathcal{A}_m \varsigma^{-2(m-1)}. \tag{10.7}$$

The derivatives of B keep its terminal argument y fixed.

Proof. Use $J_q = U_q/(Lq!)$. Grouping labeled subsets of the same size in (10.4) gives

$$\mathcal{D} J_q = (q+1)\nabla^* J_{q+1} + 2l\rho \sum_{a=1}^{q} (q-a+1)\, \mathrm{Av}_a(J_a * J_{q-a+1}), \qquad [\mathcal{D}, \nabla^*]V = l\rho \nabla^*(J_1 * V).$$

Each Av_a is the average over the labeled subsets of size a, and so has coefficient mass one. Material differentiation obeys the ordinary product rule for contractions and permutations.

We spell out the order count. After u differentiations of a jet of order q, the sum of leaf orders is at most $q+u$, the number of leaves is at most $u+1$, and the number of operation nodes is at most $2u$. The last bound allows the adjoint followed by a contraction created by a commutator. More precisely, the total expression size increases by at most three at a differentiation: replacing a leaf by its principal adjoint term costs two, replacing it by a product costs three, and commuting past an adjoint inserts J_1 and a contraction, also costing three. Thus size is at most $q+1+3u$ and weight at most $q-1+2u$. Scalar factors consist of powers of l and polynomials in ρ of degree at most u. Since $l, \rho \leqslant 1$, these factors have absolute value at most one before accounting for their differentiation coefficients.

At the next differentiation, the sum of all leaf replacement masses is at most $C\sum_i (q_i+1)^2 \leqslant C(q+2u+1)^2$. Commuting across one of at most $2u$ operation nodes and differentiating a scalar power contribute at most Cu. Hence the coefficient mass after u steps is bounded by

$$\prod_{v=0}^{u-1} C(q+2v+1)^2 \leqslant [C(q+u+1)]^{C(q+u+1)}.$$

This also counts the terms generated by product rules. Beginning with $\mathcal{D}M = \nabla^* U_1$, the exterior adjoint is preserved by the commutator identity. After $m-1$ further derivatives all terms have size $O(m)$ and weight at most $2m-1$, with a common factor L. Apply lemma 10.2 to each whole vector expression, and sum its coefficient masses. This proves (10.6).

For the Jacobian write $B' = BC$, where $C = r\rho U_1(\rho, Y_\rho)$ and $Y_\rho = T^{-1}_{\rho \to b_*}(y)$. The backward path satisfies $Y'_\rho = -rM_\rho(Y_\rho)$, so

$$C^{(j)} = r\big[\rho \mathcal{D}^j U_1 + j\mathcal{D}^{j-1} U_1\big],$$

with the second summand absent when $j = 0$. Repeated product rules express $B^{(m)}$ as a sum of ordered words $BC^{(j_1)} \cdots C^{(j_a)}$ with $\sum_i (j_i+1) = m$. Their total coefficient mass is at most $m!$: a word of order m has at most $m+1$ differentiation targets. Expand each material factor as above and contract the resulting matrices into a single expression, without using Hölder on the separate factors. Its size is at most

$$\sum_i (2+3j_i) + (a-1) = 3m-1,$$

and its weight is at most $2\sum_i j_i \leqslant 2(m-1)$. The physical factors give $l^a \leqslant l$. All word coefficients and material expansion coefficients have total mass $[C(m+1)]^{C(m+1)}$, since the allocated orders sum to at most m. The second-moment lemma bounds this single expression, while the remaining left factor has $\|B\| \leqslant e$. Finally $Y_\rho \sim \xi_\rho$ under the inverse probability transport, and conditional Gaussian isometry gives $\mathbb{E}_G |B^{(m)} G|^2 = \|B^{(m)}\|^2_{\mathrm{HS}}$. These facts prove (10.7). □

10.3 Stationary derivatives from the same calculus

Corollary 10.5 (Stationary vector observables)**.** *Suppose a complete stationary flow has generator $\mathcal{L} = \nabla^* A \nabla$ under the law of the adjoint-calculus lemma, with $A^T = -A$. Let $\alpha > 0$, and suppose $B = A/\alpha$ is a coefficient of size two and weight zero. If a vector-valued observable satisfies, for $j \geqslant 1$ and every split,*

$$|\nabla^j \phi|_{\text{split}} \leqslant a c^{j+1} j! R^{-j},$$

then, for $k \geqslant 1$,

$$\left\| \frac{d^k}{dt^k} \phi(Z_t) \right\|_2 \leqslant a\sqrt{D}\, \mathcal{A}_k (\alpha R^{-2})^k. \tag{10.8}$$

No zero-order bound on ϕ or on the score is required.

Proof. The field $F = \nabla\phi/a$ satisfies the coefficient bound with size two and weight one. Define, with contraction slots chosen in the order of the generator,

$$E_1 = \nabla^*(B * F), \qquad E_{k+1} = \nabla^*(B * \nabla E_k).$$

Skew symmetry cancels the second-order part of $\nabla^* A \nabla$, so it is the first-order generator of the stated flow. Induction gives $\mathcal{L}^k \phi = a\alpha^k E_k$. The expression grammar permits these formulas: ∇E_k has two slots; the contraction with B retains one slot in each operand; the final adjoint leaves vector output. The initial size is six and each step adds five, hence $n(E_k) = 5k + 1$. The initial weight is two and each step adds two, hence $w(E_k) = 2k$. Apply the second-moment lemma directly to E_k. The expression is kept intact until that lemma is applied, so there is no separate moment allocation or product estimate. Stationarity identifies $\|\mathcal{L}^k \phi\|_{L^2(\mu)}$ with the fixed-time trajectory derivative norm, proving the assertion. □

For the centering flow the skew block matrix in (9.5) has $\alpha = L/s$. The spatial factorial bounds for K/L verify the hypothesis on B. The score of $\xi_* \otimes \gamma_d$ has Hessian $\operatorname{diag}(\nabla^2 H_*, I)$, which has the same factorial coefficient bounds. The joint state, the frozen conditional mean, and $T - \mathrm{id}$ meet the observable hypothesis with amplitudes $a = C, CL, Cl$, respectively, by the jet and transport bounds; the corresponding dimension is at most $2d$.

We also give the conditional Gaussian application used for kernel actions. In physical-displacement time z, write

$$W_z = C + \cos(z/q)(Y - C) + q \sin(z/q) G, \qquad Y = C + qU,$$

where, conditionally on C, the vectors U, G are independent standard Gaussians, and $R/\sqrt{2} \leqslant q \leqslant 1$. The pair $(\cos(z/q)U + \sin(z/q)G, -\sin(z/q)U + \cos(z/q)G)$ is a stationary Gaussian rotation with constant skew generator of magnitude $1/q$. For $\Phi = T_\rho - \mathrm{id}$ use the observable $\Phi_C(u, g) = \Phi(C + qu)$ and set $S_0 = \min\{1, R/q\}$. The transport derivative bound implies

$$|\nabla^j \Phi_C|_{\text{split}} \leqslant Clq\, c^{j+1} j! S_0^{-j}, \qquad j \geqslant 1.$$

Indeed its direct bound is $lq^j C^{j+1} j! R^{1-j}$; when $q \leqslant R$ this is bounded by the displayed expression with $S_0 = 1$, and when $q \geqslant R$ it is bounded with $S_0 = R/q$. Apply the stationary corollary under the conditional Gaussian law. Its constants do not depend on C, so integration over C yields

$$\left\| \frac{d^m}{dz^m} \Phi(W_z) \right\|_2 \leqslant l\sqrt{d}\, \mathcal{A}_m q^{1-m} \max\{1, q/R\}^{2m} \leqslant l\sqrt{d}\, \mathcal{A}_m R^{-2m}. \tag{10.9}$$

For the last step, if $q \geqslant R$ the additional factor is $q^{m+1} R^{-2m} \leqslant R^{-2m}$; if $q < R$ use $q^{1-m} \leqslant 2^{(m-1)/2} R^{1-m} \leqslant 2^{m/2} R^{-2m}$ and absorb the exponential factor in $\mathcal{A}_m$. Since $W_0' = G$, a value interpolation rule differentiated at zero, at physical scale ψ, has RMS derivative residual $\mathcal{A}_m l\sqrt{d}\, \psi^{m-1} R^{-2m}$

and coefficient mass C_m/ψ. This follows by subtracting the Taylor polynomial of degree $m-1$ before applying the rule; its remainder has RMS at most $\psi^m \sup_z \|\Phi(W_z)^{(m)}\|_2/m!$ and the derivative rule costs C_m/ψ. Probability transport, stationary centering, and conditional Gaussian rotation therefore use the same envelope.

11 Affine collocation of stationary flows

All quadrature data are deterministic. The next section evaluates one shared affine graph and compares every stored layer directly with an exact stationary benchmark. Its only dynamical ingredients are probability flow, Gaussian rotation, and the stationary centering flow.

11.1 One conditional-mean collocation system

The probability and stationary flows define one affine system in their physical conditional-mean centers. Its numerical construction follows [Ope26, Section 6]; simultaneous evaluation uses the global Picard principle of [CCLZ26b, Appendix E]. Fix the deterministic parameters

$$R = \delta^t, \quad b_* = (1-R^2)^{1/2}, \quad \psi = \delta^{4t}, \quad h_0 = \delta^w, \quad s = \sigma/2, \qquad 0 < w < t < \frac{1}{160(K+2)}. \tag{11.1}$$

Write $l = \lambda r^2$, $L = \lambda r$, and assume

$$\max\{l, L/\sigma\} \leqslant S\delta^b, \qquad b \geqslant \tfrac{1}{2}; \tag{11.2}$$

where S is the scalar majorant in section 12; for sampling only the bound on l is required. Take $m \geqslant 2$, $R, h_0 \leqslant 1/2$, and $R \leqslant \tau \leqslant 1$.

Value quadrature. Use the m Gauss–Legendre nodes x_j, positive weights ω_j, and cardinal polynomials ℓ_j. Polynomial reproduction gives

$$a_j = \int_0^1 g\ell_j, \qquad \sum_j a_j^2/\omega_j \leqslant \|g\|_2^2, \qquad \sum_j |a_j| \leqslant \|g\|_2.$$

Thus a partial integration row with kernel bounded by one has absolute mass at most the length of its full interpolation cell. Summing the completed-cell rows and the current partial-cell row bounds every probability integration row by its full flow horizon. Taylor subtraction bounds its RMS defect by $2\ell^{m+1}D_m/m!$, where $D_m = \sup_{v\in I}\|f^{(m)}(v)\|_2$ on the deterministic cell. The Legendre reproducing kernel gives $\sup_x \sum_j |\ell_j'(x)| \leqslant m^3$, so angle differentiation on $[0,\psi]$ has row mass at most m^3/ψ. Probability meshes have equal steps at most h_0 in $-\log(1-\rho)$. Backward rows use oriented integration weights. Source nodes and endpoints are state targets; each forward or conditional probability interval has its own mesh. The angle interval $[0,\psi]$ and centering interval $[0,1]$ use the same GL rule.

For the kernel put $f(\rho) = (B_\rho - I)G$. The exact identities $f'(0) = 0$ and $f(b_*) = 0$ give the ordinary value rule

$$\begin{aligned} \int_0^{b_*} \frac{f'(\rho)}{\rho}\, d\rho &= -\frac{f(0)}{b_*} + \int_0^{b_*} g(\rho)\, d\rho, \\ g(\rho) &= \frac{f(\rho)-f(0)}{\rho^2} = \int_0^1 (1-v) f''(v\rho)\, dv, \\ \mathcal{Q}f &= -\frac{f(0)}{b_*} + \sum_{c,j} \frac{w_{cj}[f(\rho_{cj}) - f(0)]}{\rho_{cj}^2}. \end{aligned} \tag{11.3}$$

If $\|f^{(m+2)}(\rho)\|_2 \leqslant A_{m+2}(1-\rho^2)^{-m-2}$, then $1-\rho^2v^2 \geqslant \max\{1-\rho^2, 1-v\} \geqslant (2-\rho^2-v)/2$ yields

$$\|g^{(m)}(\rho)\|_2 \leqslant 2^{m+2}A_{m+2}\int_0^\infty \frac{u\,du}{(1-\rho^2+u)^{m+2}} = \frac{2^{m+2}A_{m+2}}{m(m+1)}(1-\rho^2)^{-m}.$$

Geometric cell lengths cancel this factor in Taylor's bound, so $\|\int f'/\rho - \mathcal{Q}f\|_2 \leqslant C^m A_{m+2}h_0^m/m!$. The value coefficients $\mathfrak{q}_\rho$ of $\mathcal{Q}$ satisfy

$$\sum_\rho |\mathfrak{q}_\rho| \leqslant 40m^2/h_0. \tag{11.4}$$

Indeed $\sum_j \omega_j/x_j^2 = 2m(m+1)$: apply Gaussian exactness to $p(x)^2$, where $p(x) = [1-(-1)^m P_m(2x-1)]/x$, and integrate by parts to obtain $\int p^2 = 2m(m+1)$. On subsequent cells GL quadrature of ρ^{-2} is bounded by its integral. The first-cell length is at least $h_0/4$, proving the displayed mass bound.

The affine map. Write $M_\rho(v) = u_{x+r\rho v, r\sqrt{1-\rho^2}}$ and $M_0 = u_{x,r}$. At centering nodes a, b set

$$y_a = y_0 + \sum_b w_{ab}V_b, \qquad g_a = G_0 + s^{-1}\sum_b w_{ab}[M_*(y_b) - M_0].$$

For each centering node b and kernel node ρ, put $v_\rho = 1-\rho^2$, $q_\rho = \sqrt{v_\rho/(1+\rho^2)}$, $\eta_e = 1/\sqrt{2}$, and supply an independent standard Gaussian $\zeta_{b\rho}$. Backward and conditional probability states satisfy

$$\begin{aligned}
B_{b,a} &= y_b - r\sum_j w^{\text{back}}_{a,j} M_{a_j}(B_{b,a_j}),\\
Z_{b,\rho,\eta} &= \zeta_{b\rho} - r\sqrt{v_\rho}\sum_j w^{\text{cond}}_{\eta,j} M^{\text{cond}}_{b,\rho,\eta_j}(Z_{b,\rho,\eta_j}),\\
M^{\text{cond}}_{b,\rho,\eta}(z) &= u_{x+r\rho B_{b,\rho}+r\sqrt{v_\rho}\eta z, r\sqrt{v_\rho}\sqrt{1-\eta^2}}.
\end{aligned}$$

The conditional interval is $[0, \eta_e]$. Define the explicit rotation and the subsequent forward probability states by

$$\begin{aligned}
H_{b\rho} &= \rho B_{b,\rho} + \sqrt{v_\rho}Z_{b,\rho,\eta_e}/\eta_e, & C_{b\rho} &= \rho(H_{b\rho} + \rho B_{b,\rho})/(1+\rho^2),\\
W_{b\rho z} &= C_{b\rho} + \cos(z/q_\rho)(B_{b,\rho} - C_{b\rho}) + q_\rho \sin(z/q_\rho)\, g_b,\\
F_{b,\rho,z,a} &= W_{b\rho z} - r\sum_j w^{\text{forw}}_{a,j} M_{a_j}(F_{b,\rho,z,a_j}),\\
V_b &= s^{-1}\sum_\rho \mathfrak{q}_\rho \sum_z d_z \sum_j w^{\text{forw}}_{b_*,j} M_{a_j}(F_{b,\rho,z,a_j}).
\end{aligned}$$

Now $W_0 = B$ and $W_0' = G$, so the last row uses $\partial_z(T_\rho - I)(W_z)|_0 = (B_\rho - I)G$; its factor $1/r$ cancels against the forward correction. Eliminate free states in the order centering, backward, conditional, rotation, forward, stopping at every mean value. This gives

$$\Phi(E) = b + AU(E), \qquad U(E)_i = u_{E_i,r_i}, \qquad b_i = x + r(\ell_i y_0 + m_i G_0 + n_i \zeta_{b\rho}). \tag{11.5}$$

For ordinary backward nodes $(\ell_i, m_i, n_i) = (a, 0, 0)$. Conditional nodes have $(\rho, 0, \sqrt{v_\rho}\eta)$; forward nodes have

$$(\ell_i, m_i, n_i) = a\big(1 - q_\rho^2(1-\cos(z/q_\rho)),\ q_\rho\sin(z/q_\rho), \rho\sqrt{2v_\rho}(1-\cos(z/q_\rho))/(1+\rho^2)\big).$$

All mean-node radii are at least $rR/\sqrt{2}$, and their position-anchor coefficients satisfy $0 \leqslant \ell_i \leqslant b_*$. Sampling nodes have $a \leqslant \rho_e = (1+\tau^2/4)^{-1/2}$ and $r_i \geqslant rR/\sqrt{5}$. The final rows are $O_M(E) = sG_0 + \sum_i \omega_i u_{E_i,rR}$ and $O_S(E) = y_0/\rho_e - (r/\rho_e)\sum_j w_j u_{E_j,r_j}$. The inline conditional correction carries $rv_\rho(1-\cos(z/q_\rho)) \leqslant rz^2$, since $v_\rho/q_\rho^2 = 1+\rho^2 \leqslant 2$. Thus the declared center and final row masses are

$$A_M = 256m^5(r/\sigma)\delta^{-4t-w} + 4r^2, \quad A_S = r^2, \qquad A_{f,M} = 1, \quad A_{f,S} = r. \tag{11.6}$$

There are at most $P = 16m\delta^{-w}(1+|\log R|)$ probability nodes. One shared mean vector contains M_0, m centering means, mP backward means, mP^2 conditional means, and m^2P^2 forward means. Consequently

$$N \leqslant 1+m+mP+mP^2+m^2P^2 \leqslant 1280m^4(1+|\log R|)^2\delta^{-2w}, \qquad \|A\|_{\mathrm{op}} \leqslant \sqrt{N}\max_i \sum_j |a_{ij}|. \tag{11.7}$$

Free-state endpoints add no mean coordinates. A layer evaluates each coordinate once. The final mean row has m distinct children; the sampling-only system and its final row each have at most P children. For query-preserving shifts the normalized direct-center coefficients are $k_i = 1-\ell_i/b_*$ for means and $k_i = 1-\ell_i\rho_e$ for sampling. Writing $\tau_i = r_i/r$, their ratios satisfy $|k_i|/\tau_i \leqslant 3$: use $1-a/b_* \leqslant 2\sqrt{1-a^2}$, $q_\rho^2(1-\cos(z/q_\rho)) \leqslant \psi^2/2$, and the stated radii. Also $\psi/q_\rho \leqslant \sqrt{2}R^3$. All auxiliary $\zeta_{b\rho}$ stay unshifted. A final mean child still has radius rR, $\ell_i = b_*$, and $k_i = 0$.

The exact stationary benchmark. Use exact centering, backward, conditional, and forward probability flows. The rotation preserves the required marginal: equivalently draw $A \sim \nu_{x,r}$, $B = \rho A + \sqrt{v_\rho}N$, and $H = A + \sqrt{v_\rho}N'$ with independent standard Gaussians. Then $(B-C)/q_\rho$ is standard Gaussian independent of C by orthogonal projection. The conditional flow generates exactly $H \mid B$, so its rotation with the independent momentum preserves $B \sim \xi_\rho$. For conditional nodes, Gaussian observation composition gives a center $x + r\theta Y_\theta$ and radius $r\sqrt{1-\theta^2}$, where $\theta^2 = \rho^2 + v_\rho\eta^2$ and $Y_\theta \sim \xi_\theta$. All other marginal identities follow from propositions 8.3 and 9.3; branch independence is unnecessary. Write E^* for these physical centers and use the norm $\|E\|_{\infty,2} = \max_i \|E_i\|_2$. The local integral equations and (9.6) give

$$\|E^* - b\|_{\infty,2}/r \leqslant C[(L/\sigma)\sqrt{d} + rD_F(x)] \leqslant CD_F(x), \qquad \max_i \|u_{E_i^*,r_i}\|_2 \leqslant CD_F(x). \tag{11.8}$$

The second bound is conditional Jensen. Every comparison center is a Gaussian observation of the same posterior; by lemma 6.1 and Lipschitz continuity, $\max_i \|D_F(E_i^*)\|_2 \leqslant (1+C\lambda)D_F(x)$. Use (10.6) for probability paths and (10.7) at order $m+2$ for kernel values. For angle derivatives apply corollary 10.5 to Gaussian rotation at speed $1/q_\rho$ and $\phi_C(w) = (T_\rho - I)(C + q_\rho w)$. Its amplitude is lq_ρ and radius $S = \min\{1, R/q_\rho\}$. Since $R/\sqrt{2} \leqslant q_\rho \leqslant 1$,

$$q_\rho^{1-m}S^{-2m} = \max\{q_\rho^{1-m}, q_\rho^{m+1}R^{-2m}\} \leqslant 2^{m/2}R^{-2m}.$$

Thus the physical-time derivative bound is $l\sqrt{d}\,\mathcal{A}_m R^{-2m}$ after enlarging its absolute envelope. For centering use the same lemma with $\alpha = L/s$ and amplitudes $L, 1$ for its observable and joint state. Taylor subtraction gives

defect source	power of δ
probability and kernel value quadrature	wm
conditional/forward quadrature under kernel and angle rows	$w(m-1)-4t$
Gaussian angle differentiation	$2tm-4t-w$
centering state quadrature	$(m+1)/3$

The second row uses only the weights $\psi^{-1}h_0^{-1}$; the third uses the physical-time derivative bound above. The last uses $(L/s)R^{-2} \leqslant \delta^{1/3}$. Each exponent is at least $\beta = w(m-1) - 4t$. Free-state elimination has fixed length and preserves the physical factor r. Thus one analytic envelope with $\log A_{\mathrm{num}} \leqslant C(m+2)\log(m+2)$ gives

$$\|E^* - \Phi(E^*)\|_{\infty,2} \leqslant A_{\mathrm{num}} r D_F(x)\delta^{\beta}. \tag{11.9}$$

The final mean quadrature has normalized defect at most $A_{\mathrm{num}}(L/\sigma)\sqrt{d}[(L/s)R^{-2}]^m$; sampling has defect at most $A_{\mathrm{num}}D_F(x)\delta^{wm}$. Hence, on the same anchors,

$$\|O_M(E^*) - u - sG_1\|_2 \leqslant A_{\mathrm{num}}\sigma D_F(x)\delta^{\beta}, \qquad \|O_S(E^*) - Y_{\rho_e}/\rho_e\|_2 \leqslant A_{\mathrm{num}} D_F(x)\delta^{\beta}.$$

Here $G_1 \sim \gamma_d$ and $Y_{\rho_e}/\rho_e \sim \nu_{x,r} * \mathcal{N}(0, \tau^2 I_d/4)$ by the exact stationary and probability-flow identities. These are the entire numerical inputs to the shared-layer Gaussian compiler.

12 Recursive evaluation by original gradients

We execute the affine graph of section 11.1 once per synchronous layer. A single recursive mean routine evaluates its nodes; a sampling block is a finite graph of mean calls. The construction adapts Gaussian absorption from [Ope26, Section 7]. Its distinguishing point is that Gaussian completion is performed jointly across a stored layer, so a node value can be used by several rows without duplicating its computation.

Throughout, $g = \nabla F$, $\mathrm{Lip}(g) \leqslant \lambda = \delta$, $L = \lambda r$, $l = \lambda r^2$, and $D_F(x) = \sqrt{d} + |g(x)|$. Fix $K \geqslant 2$, $40(K+2)t < 1/4$, $0 < w < t$, and $R = \delta^t \leqslant 1/2$, $b_* = (1-R^2)^{1/2}$, $D_s = \delta^{-t}$. Every Gaussian tape consists of finitely many d-dimensional standard Gaussian blocks. Matrices acting on the block index implicitly tensor with I_d. Counts below concern gradient queries and the number of these Gaussian blocks; matrix arithmetic is not an oracle query.

12.1 Gaussian absorption and a joint layer coupling

Lemma 12.1 (Gaussian completion)**.** *Suppose a finite mean program, with designated input P, has the query-preserving identity*

$$Q(x+\Delta, P; \zeta + v\otimes\Delta/r) = Q(x, P; \zeta), \qquad |v| \leqslant D_s. \tag{12.1}$$

The shift vector v depends on scalar program data, not on x, P, or oracle answers. If $0 < \rho \leqslant r/(2D_s)$, put $u = \rho v/r$, $a = \sqrt{1-|u|^2}$, and $T_v = (I - uu^{\mathsf{T}})^{1/2}$. For independent standard Gaussian Z, ζ, independent also of P, set $H = T_v\zeta - u\otimes Z$. Then H is standard Gaussian and independent of P, and

$$Q(x+\rho Z, P; T_v\zeta) = Q(x, P; H)$$

with identical oracle queries. Moreover $\|T_v^{-1}\| \leqslant 2$ and $T_v^{-1}v = v/a$.

Proof. Since $T_v u = au$, the matrix

$$\begin{pmatrix} T_v & -u \\ u^{\mathsf{T}} & a \end{pmatrix}$$

is orthogonal. Applied to (ζ, Z), its first component is H. This proves its Gaussian law and independence of any random variable independent of (ζ, Z). Taking $\zeta' = T_v\zeta - v\otimes\rho Z/r$ in (12.1) proves the pathwise identity. The eigenvalue of T_v in the direction v is $a \geqslant \sqrt{3}/2$; all perpendicular eigenvalues are one. This proves both inverse bounds. All assertions hold conditionally on a sigma-field whenever the center and scalar data are measurable there and the stated Gaussian independence holds conditionally. □

The executable layer. Write the graph as $E = b + \mathsf{A}U(E)$, where $U(E)_i = u_{E_i,r_i}$, there are n_c coordinates, and $A > 0$ is a declared bound on every absolute row sum of A. Its radii satisfy $r_i \geqslant rR/\sqrt{5}$. Define

$$\rho = \frac{rR}{2\sqrt{5}D_s}, \qquad \Lambda = \sqrt{n_c}A, \qquad \sigma_c = \frac{\rho}{2\Lambda}. \tag{12.2}$$

For the child at node i, whose auxiliary shift is v_i, set

$$T_i = (I - \rho^2 v_i v_i^{\mathsf{T}}/r_i^2)^{1/2}. \tag{12.3}$$

These matrices are determined bottom-up from finite child programs. They are not selected from the values of the current centers.

The actual direct vector may be denoted $\widehat{b}$ when it uses an approximate private anchor. Initialize $\widehat{E}^{[0]} = \widehat{b} + \rho Z^{[0]}$, with independent Gaussian coordinate blocks. At layer j, draw fresh independent inputs $P_i^{[j]}$ and raw auxiliary tapes $\zeta_i^{[j]}$, and evaluate

$$Q_i^{[j]} = \mathsf{M}_i(\widehat{E}_i^{[j]}, P_i^{[j]}; T_i\zeta_i^{[j]}).$$

Store these values and use them in every required row. Independently of all these tapes, draw a standard Gaussian array $Z^{[j+1]}$ and put

$$\widehat{E}^{[j+1]} = \widehat{b} + \mathsf{A}Q^{[j]} + CZ^{[j+1]}, \qquad CC^{\mathsf{T}} = \rho^2 I - \sigma_c^2 \mathsf{A}\mathsf{A}^{\mathsf{T}}. \tag{12.4}$$

Indeed $\|\mathsf{A}\|_{\mathrm{op}} \leqslant \sqrt{n_c}A$, by the Frobenius bound applied row by row. Hence $CC^{\mathsf{T}} \succeq 3\rho^2 I/4$. The fixed positive square root is well defined. The correlation of this residual across nodes is part of the algorithm.

What a retained child coupling supplies. At a deterministic center z, a child accuracy statement means the following joint construction. Its actual output is the deterministic program $Q_i(z, P_i; H_i)$, where (P_i, H_i) has product standard Gaussian law. On an extension of this probability space one can draw P_i' such that

$$\mathsf{Law}(P_i, P_i') = \gamma_d(dP_i)\, Q_i^{\mathrm{ref}}(z; P_i, dP_i'), \qquad P_i' \sim \gamma_d,$$

and the RMS error from $u_{z,r_i} + \sigma_i P_i'$ has the stated bound. Here Q_i^{ref} is the prescribed exact kernel, not an arbitrary Gaussian marginal coupling. Disintegrate this construction with respect to (P_i, H_i) to obtain a conditional kernel for P_i'. Applying that kernel to another pair with the same product law preserves the full retained coupling. For several children use the product of these conditional kernels. This is legitimate on the finite-dimensional Euclidean tape spaces; equivalently, each construction can use its own additional independent uniform random variable. These variables occur only in the proof and are not program inputs.

Lemma 12.2 (The shared-layer induction)**.** *Let $\mathcal{A}$ be an outside sigma-field. Conditional on $\mathcal{A}$, fix $\widehat{b}$, the exact benchmark E^*, the graph, and all scalar data. Suppose each child has fixed-tape center Lipschitz constant $U\lambda$ and retained error at center z at most $B\sigma_c\delta^\alpha D_F(z)$. There is a joint construction of all actual layers and comparison arrays*

$$\widetilde{E}^{[j]} = E^* + \rho\widetilde{Z}^{[j]}, \qquad \mathsf{Law}(\widetilde{Z}^{[j]} \mid \mathcal{A}) = \gamma_{n_c d}.$$

The actual layers have exactly the law of the executable program. Writing $e_j = \max_i \|\widehat{E}_i^{[j]} - \widetilde{E}_i^{[j]}\|_2$ and $\overline{D} = \max_i \|D_F(E_i^*)\|_2$, one has $e_0 = \max_i \|\widehat{b}_i - E_i^*\|_2$ and

$$e_{j+1} \leqslant U\lambda A e_j + \max_i \|E_i^* - \widehat{b}_i - (\mathsf{A}U(E^*))_i\|_2 + \frac{B\rho}{2}\delta^\alpha \overline{D}. \tag{12.5}$$

The reference noises at different layers need not be independent. No Gaussian law is asserted for the actual centers.

Proof. At layer zero use $\widetilde{Z}^{[0]} = Z^{[0]}$. This proves the initial assertion. Assume the coupling has been constructed through layer j. Draw the fresh executable inputs and tapes independently of everything constructed so far, and keep the program output at its actual center. On the same tapes form the intermediate output

$$Q_{i,0}^{[j]} = Q_i(E_i^* + \rho\widetilde{Z}_i^{[j]}, P_i^{[j]}; T_i\zeta_i^{[j]}).$$

The deterministic child Lipschitz estimate gives, for each i,

$$\|Q_i^{[j]} - Q_{i,0}^{[j]}\|_2 \leqslant U\lambda e_j.$$

Conditional on $\mathcal{A}$, the triples $(\widetilde{Z}_i^{[j]}, P_i^{[j]}, \zeta_i^{[j]})$ are independent across i, with product standard Gaussian law. Therefore absorption identifies $Q_{i,0}^{[j]}$ pathwise with $Q_i(E_i^*, P_i^{[j]}; H_i^{[j]})$, where

$$H_i^{[j]} = T_i\zeta_i^{[j]} - (\rho/r_i)v_i \otimes \widetilde{Z}_i^{[j]}.$$

The pairs $(P_i^{[j]}, H_i^{[j]})$ are independent product Gaussians conditional on $\mathcal{A}$. Use the product disintegration just described to draw reference outputs $u_{E_i^*,r_i} + \sigma_c G_i^{[j]}$. Conditional on $\mathcal{A}$, the $G_i^{[j]}$ are independent standard Gaussians. Integration of the conditional child estimates gives

$$\|Q_{i,0}^{[j]} - u_{E_i^*,r_i} - \sigma_c G_i^{[j]}\|_2 \leqslant B\sigma_c\delta^\alpha \|D_F(E_i^*)\|_2.$$

Now draw the fresh executable residual $Z^{[j+1]}$ independently of this entire coupling, and define

$$\rho\widetilde{Z}^{[j+1]} = \sigma_c \mathsf{A}G^{[j]} + CZ^{[j+1]}, \qquad \widetilde{E}^{[j+1]} = E^* + \rho\widetilde{Z}^{[j+1]}.$$

Its conditional law is centered Gaussian with covariance $\rho^2 I$. A Gaussian with this covariance has independent coordinate blocks, which proves the next-layer invariant. Subtracting this display from (12.4) gives the exact identity

$$\widehat{E}^{[j+1]} - \widetilde{E}^{[j+1]} = \widehat{b} + \mathsf{A}U(E^*) - E^* + \mathsf{A}\{Q^{[j]} - U(E^*) - \sigma_c G^{[j]}\}.$$

Apply Minkowski separately to each row. Its first child error is at most $U\lambda A e_j$ and its second at most $BA\sigma_c\delta^\alpha\overline{D} \leqslant B\rho\delta^\alpha\overline{D}/2$. This proves the recurrence.

The construction only augments the executable variables. Every new program tape is still sampled independently of the previous program tapes, and the program never reads a reference variable. Finally, the conditional Gaussian assertion was used only given $\mathcal{A}$. Conditioning additionally on the complete execution history would in general destroy it, and is not part of the induction. □

Fresh final rows, including retained inputs. Suppose a final row uses distinct graph coordinates with coefficients a_i, $\sum_i |a_i| \leqslant A_f$, and new child means with widths σ_i. The layer coupling is constructed without using their designated inputs P_i, their auxiliary tapes, or an optional comparison coordinate G_E. Assume that, conditional on $\mathcal{A}$, the P_i and G_E are independent standard Gaussians and independent of that coupling. At the final centers use fresh tapes ζ_i and transform them by (12.3), using each final child's own shift and radius. The proof of the layer lemma now applies once, with (P_i, G_E) retained. It gives final reference inputs P_i' with

$$\mathsf{Law}((P_i, P_i')_i, G_E \mid \mathcal{A}) = \bigotimes_i \{\gamma_d(dP_i)Q_i^{\mathrm{ref}}(E_i^*; P_i, dP_i')\} \otimes \gamma_d(dG_E),$$

and the row error is at most

$$\sum_i |a_i| \left[U\lambda e_{N_{\mathrm{P}}} + B\sigma_i \delta^{q_i} \|D_F(E_i^*)\|_2 \right]. \tag{12.6}$$

The formula follows because the absorbed auxiliary tapes are independent of all the designated inputs and of G_E. The reference kernels are then sampled independently using their disintegrations. They need not be independent of the actual center errors.

For an ordinary sampling row, take independent designated inputs, choose $\sigma_i \leqslant n/(2A_f)$, and add a fresh Gaussian residual of variance $\beta^2 = n^2 - \sum_i a_i^2\sigma_i^2 \geqslant 3n^2/4$. The reference noise $\sum_i a_i\sigma_i P_i' + \beta G_E$ is then Gaussian of variance $n^2 I$, independent of the outside anchors. For a retained mean row, the same identity is obtained from the local Gaussian input assembly; G_E is only a comparison coordinate and no extra residual is drawn. We verify its hypotheses for that assembly below.

12.2 One recursive mean and one finite sampling block

The mean program is $\mathsf{M}^q_{x,r,\sigma}(P;\zeta)$. Only its precision q is a recursive label. A public mean root has $q = K$ and $\max\{l, L/\sigma\} \leqslant \delta$. For the size proof, attach to a reachable mean a certificate $(b, \mathfrak{s})$ satisfying

$$\max\{l, L/\sigma\} \leqslant \mathfrak{s}\delta^b, \qquad 1 \leqslant \mathfrak{s} \leqslant S. \tag{12.7}$$

The root certificate is $(1,1)$; neither certificate coordinate is a program parameter. The upper envelope S is specified below. If $q \leqslant 1/2$, draw a Gaussian block Z and return

$$\mathsf{M}^q_{x,r,\sigma}(P;Z) = g(x + rZ) + \sigma P.$$

Its auxiliary shift is -1 in the Z block and zero elsewhere.

For a nonterminal mean use the simultaneous mean graph. Put $s = \sigma/2$, $n = \sqrt{3}\sigma/2$. Its active final weights $\omega_i > 0$ have sum one, and all its active final radii are rR. The allocation is

child role	width	precision
active final mean	$Rn/2$	$q - t$
mean in a main synchronous layer	σ_c	$q - 1/4$
mean in the private sampling block	sampling allocation	$q - 1/4$.

(12.8)

In the main layers, $A = A_M$ in (12.2). The equality $R\delta^{q-t} = \delta^q$ is the reason for the smaller active final width. Each executed node in each layer is a direct child of the parent mean; sharing its value among rows creates no additional recursive call.

The private position anchor is $\widehat{Y} = b_* \mathsf{S}^{q-1/4}_{R/b_*}(x, r)$, with the block defined below. Independently of its tape, perform the local Gaussian assembly (7.1), with

$$\sigma_i = Rn/2, \qquad \beta^2 = n^2 - \sum_i \omega_i^2\sigma_i^2 \geqslant 9\sigma^2/16.$$

It supplies the main momentum anchor G_0 and designated final inputs P_i. Also draw the independent conditional-flow anchors $\zeta_{b\rho}$ required by the graph. All main synchronous layers use fresh standalone mean inputs and raw tapes. Only the final layer uses the designated P_i, which have not been read by the private or main center computations. Return

$$\mathsf{M}^q_{x,r,\sigma}(P;\zeta) = \sigma P + \sum_i \omega_i \left[\mathsf{M}^{q-t}_{\widehat{E}_i^{[N_{\mathrm{P}}]}, rR, \sigma_i}(P_i; T_i\zeta_i) - \sigma_i P_i \right]. \tag{12.9}$$

Every leaf evaluates the original gradient, and every nonterminal correction is a positive weighted average of child corrections.

For $R \leqslant \tau \leqslant 1$, the block $\mathsf{S}^{\alpha}_{\tau}(x, r)$ uses the sampling-only graph, Gaussian position anchor Y_0, $A = A_S = r^2$, and $n = \tau/\sqrt{2}$. All layer means have precision α and width σ_c. At the final row use fresh means of precision α and width $n/(2A_f)$, where $A_f = r$. Its direct term is Y_0/ρ_e, $\rho_e = (1 + \tau^2/4)^{-1/2}$. Add the residual with variance $n^2 - \sum_i a_i^2\sigma_i^2$, followed by an independent reserve $\tau G/2$. The output before this last reserve is denoted $\mathsf{S}^{\alpha}_{\tau,\mathrm{pre}}$. There is no recursive sampling call and no terminal sampling rule. The private block has $\alpha = q - 1/4$ and $\tau = R/b_* \in [R, 1]$. The public decoder block has $(\tau, \alpha) = (1, K)$ and $l \leqslant \delta$.

12.3 Finite realization and deterministic bounds

Choose $B_0 \geqslant 1$ to dominate the numerical coefficient polynomials and every constant in a single size transition, including their products with the polynomial factor in $\sqrt{n_c} \leqslant B_0\delta^{-w}$. Choose any integer $H \geqslant 2 + \lceil (K - 1/2)/t \rceil$ and put

$$S = (32B_0)^{H+2}.$$

and impose, for a sufficiently large absolute C_*,

$$C_* H B_0 S \delta^{1/4} \leqslant 1. \tag{12.10}$$

Increasing C_* once will also pay the finitely many absolute constants in the accuracy estimates. The schedule in section 13 pays this test and the separate deterministic shift budget.

Lemma 12.3 (Finite deterministic realization)**.** *Every reachable mean has $b > 3/4$ and $\mathfrak{s} \leqslant S$, and*

$$\lambda A \leqslant 2B_0 S\delta^{2/3} =: \eta, \qquad 2\lambda A \leqslant \delta^{1/3}. \tag{12.11}$$

Let N_c bound each local graph inventory and put $B = C(N_{\mathrm{P}} + 2)(N_c + 1)$ for a sufficiently large absolute C. Then

$$\text{work} \leqslant B^{H+1}, \qquad |v| \leqslant B^{H_s+1}, \qquad H_s \geqslant 8K + 4. \tag{12.12}$$

If $B^{H_s+1} \leqslant D_s$, all the programs are well defined on every execution. The mean shift preserves its input, output, and every query. Its fixed-tape center Lipschitz constant is at most 2λ; with the position anchor treated as a separate argument its constant is at most $3L$. The sampling block has center constant at most $5r\lambda$ and a query-preserving shift changing its output by $-\Delta/r$.

Proof. Height and size propagation. Every edge between recursive mean calls decreases q by t or $1/4$. The public sampling block adds only one nonrecursive level. Hence H bounds every path to a terminal mean. Let j count the private edges (those decreasing q by $1/4$), and set $b = 1 - 10tj$. Public sampling children may start with $j = 1$. Since there are at most $4K + 1$ subsequent private edges,

$$b \geqslant 1 - 10t(4K + 2) > 3/4.$$

An active final mean has $r_i = rR$ and $\sigma_i = \sqrt{3}R\sigma/4$; therefore $l_i = lR^2$ and $L_i/\sigma_i = (4/\sqrt{3})L/\sigma$. Its certificate keeps the same b and multiplies $\mathfrak{s}$ by at most three.

For a main-layer child, $r_i \leqslant r$ and

$$\frac{L_i}{\sigma_c} \leqslant 4\sqrt{5}\,\frac{D_s\sqrt{n_c}\lambda A_M}{R}.$$

The explicit mass $A_M = 256m^5(r/\sigma)\delta^{-4t-w} + 4r^2$ gives

$$\lambda A_M \leqslant CB_0\mathfrak{s}\delta^{b-4t-w}, \qquad \frac{L_i}{\sigma_c} \leqslant CB_0\mathfrak{s}\delta^{b-6t-2w}.$$

Here B_0 was chosen to include the product of the coefficient and inventory polynomials. Since $w < t$, the exponent is at least $b - 10t$. Also $l_i \leqslant l$. Thus a private transition costs at most a fixed multiple of B_0 in $\mathfrak{s}$ and $10t$ in b. For a private sampling block, its final mean satisfies $L_i/\sigma_i \leqslant Cl/\tau \leqslant C\mathfrak{s}\delta^{b-t}$; its layer means satisfy $L_i/\sigma_c \leqslant CD_s\sqrt{n_c}l/R \leqslant CB_0\mathfrak{s}\delta^{b-2t-w}$. These also fit $b - 10t$. The public sampling block starts from $l \leqslant \delta$ and $\tau = 1$, so its children fit $(b, \mathfrak{s}) = (1 - 10t, CB_0)$. Multiplying at most $H + 1$ transition factors proves $\mathfrak{s} \leqslant S$.

Since $b > 3/4$ and $4t + w < 1/12$, the mass formula gives $\lambda A \leqslant 2B_0S\delta^{2/3}$. The scalar test implies $2\lambda A \leqslant \delta^{1/3}$ and, after enlarging C_*, $\eta \leqslant 1/100$ and $20H\eta \leqslant 1/2$. These implications use only $\delta^{2/3} \leqslant \delta^{1/4}$ and the stronger spare factor $\delta^{1/3}$ where needed.

The query-preserving shifts. First consider a sampling block and increase its external center by Δ. Shift Y_0 by $-\rho_e\Delta/r$, holding all output-residual and reserve seeds fixed. A direct node center then changes by $k_i\Delta$, where $k_i = 1 - \ell_i\rho_e$. For a mean parent the required position-anchor displacement is $-\Delta/(b_*r)$. Obtain it as follows: apply the ordinary sampling shift to its private S block, which moves $b_*\mathsf{S}$ by $-b_*\Delta/r$, and additionally shift that block's reserve seed by $-2R\Delta/(b_*r)$. The latter changes the scaled anchor by $-R^2\Delta/(b_*r)$. Their sum is $-\Delta/(b_*r)$ because $b_*^2 + R^2 = 1$. This extra reserve displacement changes no query.

Hold the entire local Gaussian input assembly fixed, as well as every conditional-flow anchor $\zeta_{b\rho}$. In the mean graph a direct center changes by $k_i\Delta$, with $k_i = 1 - \ell_i/b_*$. The numerical coefficient bounds give $|k_i|/\tau_i \leqslant 3$, where $\tau_i = r_i/r$, for both kinds of block. In the raw auxiliary tape of each child make the shift

$$(k_i/\tau_i)T_i^{-1}v_i \otimes \Delta/r. \tag{12.13}$$

After transformation by T_i this is exactly $v_i \otimes k_i\Delta/r_i$. The child's designated input is unchanged, so its inductive identity preserves its output and every oracle query. Leave all joint center-residual seeds fixed. At the first layer this proves the required center displacements; if it holds at one layer, the shared vector of child outputs is unchanged, so the next layer's only displacement is that of its direct vector. This proves the claim through every finite layer.

A mean's final children have $\ell_i = b_*$ and $k_i = 0$. Its correction output is therefore unchanged, with its entering momentum fixed. For a sampling block the final mean values are unchanged and the direct term Y_0/ρ_e changes by $-\Delta/r$. Its residual and reserve are fixed, giving the asserted sampling shift.

Work, shift norm, and bottom-up construction. A local block has at most $C(N_{\mathrm{P}} + 2)(N_c + 1)$ direct mean calls and Gaussian blocks. This counts every private layer, the fresh final calls, and the private sampling block when present. A stored value contributes one call even if many rows use it. Consequently the work at height h is bounded by B times one plus the maximum child work, and enlarging B once yields the first bound in (12.12).

Each raw child shift in (12.13) has norm at most $6|v_i|$. Its tape is disjoint from those of the other executed children. The top-level sampling-position shift has norm at most one; the additional private reserve shift has coefficient at most $2R/b_* \leqslant 2$. Thus the squared parent shift norm is at most $C + 36\sum_i |v_i|^2$, with an absolute enlargement covering the private sampling block. Every nonzero recursive term in this sum has lost $1/4$ in precision: active final mean edges have coefficient zero. There are at most $4K + 2$ such decreases; any chosen $H_s \geqslant 8K + 4$ also counts the intervening block levels. This gives $|v| \leqslant B^{H_s+1}$. The relation $T_i^{-1}v_i = v_i/a_i$ shows that a zero coordinate block stays zero under absorption; in particular the local input-assembly slots never acquire a shift.

These statements define the programs bottom-up. Terminal shifts are known. At a parent, the already constructed child shifts obey the uniform budget D_s, so the matrices T_i exist; the above calculation then constructs and bounds the parent shift. The work and shift bounds use no center or oracle value, so they apply to every execution.

Fixed-tape center and anchor stability. All Gaussian linear transformations depend only on the scalar data. Suppose the children have center constants at most $F\lambda$, $F \leqslant 2$. For two executions of a local graph with direct vectors differing by at most d_b in each coordinate, let d_j be the maximum node displacement. The same Gaussian tapes cancel, so

$$d_0 \leqslant d_b, \qquad d_{j+1} \leqslant d_b + F\lambda A d_j, \qquad d_j \leqslant \frac{d_b}{1 - F\lambda A}.$$

The finite sum proves this bound directly, without using a fixed point. For the sampling block, changing x by Δ gives $d_b = |\Delta|$; its direct output does not depend on x. Its center constant is at most $Fr\lambda/(1 - Fl) \leqslant 5r\lambda$.

For a mean parent, the private position anchor therefore has center constant at most $Fr\lambda/(1-Fl)$, since $b_* \leqslant 1$. Every direct main center has $|\ell_i| \leqslant 1$, so its displacement is at most $[1+Fl/(1-Fl)]|\Delta|$. The final positive weights sum to one. Thus the parent's center constant, divided by λ, is at most

$$\frac{F}{1 - F\eta}\left(1 + \frac{Fl}{1 - Fl}\right) \leqslant \frac{F}{(1 - F\eta)^2} \leqslant F(1 + 20\eta),$$

where $l \leqslant \eta$, $F \leqslant 2$, and $\eta \leqslant 1/100$. Starting with the terminal factor one and using $(1 + 20\eta)^H \leqslant e^{20H\eta} < 2$ closes the induction. If only the main position anchor changes, then $d_b \leqslant r|\Delta Y|$ and the final constant is at most $2L/(1-2\eta) \leqslant 3L$. These are pathwise estimates for arbitrary fixed entering momenta and raw tapes, including the non-Gaussian transformed tapes used by a parent. □

12.4 One accuracy induction, with the input retained

Lemma 12.4 (Terminal error)**.** *A terminal mean has reference $P' = P$ and retained RMS error at most $\sigma\delta^q D_F(x)$ under (12.10).*

Proof. By lemma 6.1, $\|g(x + rZ) - u_{x,r}\|_2 \leqslant 3LD_F(x)$. Since $b > 3/4$ and $q \leqslant 1/2$,

$$\frac{3L}{\sigma\delta^q} \leqslant 3S\delta^{b-q} \leqslant 3S\delta^{1/4} \leqslant 1.$$

The actual designated input is unchanged in this construction, so this is a coupling with the identity reference kernel, not just a marginal Gaussian approximation. □

Proposition 12.5 (Finite gradient oracles)**.** *Suppose the exact stationary residual and final quadrature defects of section 11.1 are at most δ^{K+1} in their stated units, and $N_{\mathrm{P}} \geqslant 3(K + 2)$. Under the scalar and shift budgets above, every reachable mean has a coupling*

$$\mathsf{Law}(P, P') = \gamma_d(dP)Q^q_{x,r,\sigma}(P, dP'), \qquad \|\mathsf{M}^q_{x,r,\sigma}(P;\zeta) - u_{x,r} - \sigma P'\|_2 \leqslant C\sigma\delta^q D_F(x),$$

where Q^q is precisely the exact reference kernel of section 7 for its finite precision tree. The actual (P, ζ) has product standard Gaussian law. The estimate is uniform in the center and admissible scalar data; it is not a pointwise estimate conditional on $P = p$. The deterministic work, shifts, and stability apply for every fixed input and tape. For each invoked sampling block,

$$W_2(\mathsf{Law}(\mathsf{S}^{\alpha}_{\tau,\mathrm{pre}}), \nu_{x,r} * \mathcal{N}(0, 3\tau^2 I/4)) \leqslant C\tau\delta^{\alpha} D_F(x).$$

Its independent reserve gives TV error at most $C\delta^{\alpha} D_F(x)$ from $\nu_{x,r} * \mathcal{N}(0, \tau^2 I)$. Forgetting the input of a mean coupling gives its ordinary noisy-mean W_2 guarantee.

Proof. We induct on the finite recursive mean tree. Assume all proper mean children have normalized error at most $E \geqslant 1$. A private sampling block consists entirely of proper mean children at precision $\alpha = q - 1/4$; prove its guarantee first, and then prove the parent's retained mean guarantee. A public sampling block is evaluated after all of its precision-K mean children have been proved. This order contains no circular sampling induction.

Benchmark moments and the finite geometric sum. Every exact physical node in section 11.1 has a Gaussian-observation marginal $E_i^* = x + r\theta Y_\theta$, for some $0 \leqslant \theta \leqslant 1$, with $Y_\theta \stackrel{d}{=} \theta Z + \sqrt{1-\theta^2}G$, $Z \sim \nu_{x,r}$, $G \sim \gamma_d$ independent. The moment lemma and independence give $\|Y_\theta\|_2 \leqslant D_F(x)$. Hence

$$\overline{D} = \max_i \|D_F(E_i^*)\|_2 \leqslant (1 + \lambda r) D_F(x) \leqslant \tfrac{6}{5} D_F(x). \tag{12.14}$$

The last inequality is paid by the scalar test. Neither independence between different benchmark nodes nor a bound on an RMS maximum is needed. With exact anchors, (11.8) gives $e_0 \leqslant C r D_F(x)$, and the stationary residual is at most $r\delta^{K+1} D_F(x)$. Put $\chi = 2\lambda A \leqslant \delta^{1/3} \leqslant 1/10$. Iterating (12.5), rather than passing to an infinite solver, gives

$$\begin{aligned} e_{N_{\mathrm{P}}} &\leqslant \chi^{N_{\mathrm{P}}} e_0 + \frac{r\delta^{K+1} D_F(x) + (E\rho/2)\delta^\alpha \overline{D}}{1-\chi} \\ &\leqslant C r \delta^{K+1} D_F(x) + \frac{E\rho}{2(1-\chi)} \delta^\alpha \overline{D}. \end{aligned}$$

Indeed $\chi^{N_{\mathrm{P}}} \leqslant \delta^{K+2}$ by the chosen layer count.

The private sampling guarantee. For this application the outside field is generated by the Gaussian position anchor Y_0 and any fixed scalar data. Its exact benchmark is the probability-flow path from that anchor. The actual graph uses the same anchor, so there is no anchor discrepancy in the previous display. Use the fresh final-row construction with $A_f = r$, $n = \tau/\sqrt{2}$, and $\sigma_i = n/(2r)$. The reference final noise is independent of Y_0, because conditional on Y_0 all its components are independent standard Gaussians. Its variance is $n^2 = \tau^2/2$. The final exact-path row differs by at most $\delta^{K+1} D_F(x)$ from Y_{ρ_e}/ρ_e, whose marginal is $\nu_{x,r} * \mathcal{N}(0, \tau^2 I/4)$. Consequently the comparison output before the reserve has the required law with total Gaussian variance $3\tau^2/4$.

The intrinsic final mean errors sum to at most $En\delta^\alpha \overline{D}/2$. The center contribution is $2\lambda r e_{N_{\mathrm{P}}}$. Dividing all contributions by $\tau\delta^\alpha D_F(x)$, and using $\alpha \leqslant K$ and $\tau \geqslant R$, gives the explicit bound

$$\mathcal{E}_S \leqslant \frac{6}{5}\frac{E}{2\sqrt{2}} + CE\frac{l}{D_s} + C(1+l)\delta^{1-t}.$$

Here we used $\rho/r = R/(2\sqrt{5}D_s)$ and $1/(1-\chi) \leqslant 10/9$. Since $6/(10\sqrt{2}) < 0.425$, the scalar test can ensure $Cl/D_s \leqslant 1/20$ and $C(1+l)\delta^{1-t} \leqslant 1/2$. Thus

$$\mathcal{E}_S \leqslant E/2 + 1. \tag{12.15}$$

Couple the added reserve identically on both sides. The same W_2 bound therefore holds for the full block and its target $\nu_{x,r} * \mathcal{N}(0, \tau^2 I)$.

Coupling the actual and ideal private anchors. Keep the entire executable private sampling tape. By disintegration of the full-block coupling, attach to it an ideal anchor Y with law $b_* Z + RN$, while its actual scaled output remains $\widehat{Y} = b_* \mathsf{S}_{R/b_*}^{q-1/4}(x, r)$. By (12.15),

$$\|\widehat{Y} - Y\|_2 \leqslant R(E/2+1)\delta^{q-1/4} D_F(x). \tag{12.16}$$

This coupling is drawn independently of the local Gaussian input assembly and of every main-graph tape. Under standard Gaussian parent input, that assembly has the equivalent product representation

$$G_0, (P_i)_i, G_E \text{ independent standard Gaussian}, \qquad \sigma P = sG_0 + \sum_i \omega_i \sigma_i P_i + \beta G_E.$$

This representation is an equality in joint law with the executable conditional assembly; it does not resample or replace its entering P.

For the main shared-layer proof use the outside sigma-field generated by $(\widehat{Y}, Y)$, G_0, and all conditional-flow Gaussian anchors $\zeta_{b\rho}$. Do not add P, the final P_i, or G_E to this field. The exact main benchmark is constructed from Y, G_0 and those conditional-flow anchors; the executable direct vector uses $\widehat{Y}, G_0$ and the same conditional-flow anchors. Conditional on this outside field, the unused (P_i, G_E) are still independent standard Gaussians. They are independent of all main-layer tapes and of the layer coupling, whose auxiliary variables are generated without reading them.

The numerical graph has $|\ell_i| \leqslant 1$, so $\|\widehat{b} - b\|_{\infty,2} \leqslant r\|\widehat{Y} - Y\|_2$. Its exact benchmark residual relative to $\widehat{b}$ is

$$E^* - \widehat{b} - \mathsf{A}U(E^*) = \{E^* - b - \mathsf{A}U(E^*)\} + (b - \widehat{b}).$$

Thus the same layer lemma incorporates the anchor error as part of its single residual. It also changes e_0 by at most $r\|\widehat{Y} - Y\|_2$. Applying the finite geometric sum yields

$$e_{N_{\mathrm{P}}} \leqslant CD_F(x) \left[r\delta^{K+1} + rR(E+1)\delta^{q-1/4} + \rho E \delta^{q-1/4} \right]. \tag{12.17}$$

The moment bound (12.14) concerns the exact benchmark's unconditional law and is unaffected by the coupling with $\widehat{Y}$. There is no separate assertion about the distribution or moments of the actual main centers.

Final designated inputs and the exact parent kernel. Apply the fresh final-row construction to the unused designated P_i, retaining G_E. This gives independent child reference pairs (P_i, P_i') conditional on the outside field, with precisely the child kernels evaluated at the exact final centers. Those centers belong to the exact parent centering path (Y_t, G_t); they do not depend on the auxiliary conditional-flow anchors. Hence, after forgetting those anchors and the approximate private tape, the reference pair law conditional on the exact parent path is the product of the same child kernels. Define

$$\sigma P' = sG_1 + \sum_i \omega_i \sigma_i P_i' + \beta G_E.$$

The exact reference construction in section 7 uses an independent stationary position Y, the same Gaussian input assembly, and these same conditional child kernels. We have therefore constructed its full joint law (P, P'), including the prescribed entering momentum, rather than only its marginal output law.

To see the error without suppressing the cancellation, write $u_i = u_{E_i^*, rR}$. Equations (12.9) and the two input-assembly identities give

$$\begin{aligned} \mathsf{M}^q_{x,r,\sigma}(P;\zeta) - u_{x,r} - \sigma P' = \sum_i \omega_i \{\mathsf{M}_i(\widehat{E}_i, P_i; T_i\zeta_i) - u_i - \sigma_i P_i'\} \\ + \left\{ sG_0 + \sum_i \omega_i u_i - u_{x,r} - sG_1 \right\}. \end{aligned}$$

The second brace is exactly the stationary final quadrature defect of section 11.1, at most $\sigma\delta^{K+1}D_F(x)$ in RMS. The first brace is controlled by (12.6). Since $\sum_i \omega_i = 1$ and $\sigma_i\delta^{q-t} = (\sqrt{3}/4)\sigma\delta^q$, its intrinsic part, normalized by $\sigma\delta^q D_F(x)$, is at most $6\sqrt{3}E/20 < 0.520E$. By (12.17), its center part is at most

$$C\frac{L}{\sigma}\left[\delta^{K+1-q} + (E+1)R\delta^{-1/4} + E(\rho/r)\delta^{-1/4}\right].$$

Here $q \leqslant K$, $L/\sigma \leqslant S\delta^b$, $b > 3/4$, and $\rho/r \leqslant R$. The scalar test can therefore make this at most $(E+1)/20$, and also make the normalized quadrature defect at most $1/2$. Consequently

$$E_{\text{parent}} \leqslant \tfrac{5}{8}E + 1.$$

Taking $E = 4$ closes the induction, since the terminal constant is one and $(5/8)4 + 1 < 4$. The private sampling bound is then at most three; the public sampling block follows from the same calculation after all its mean children are proved.

Finally, for any coupling of pre-reserve outputs, use the same reserve on both sides and the Gaussian bound $\mathsf{TV}(\mathcal{N}(a,\tau^2 I/4),\mathcal{N}(b,\tau^2 I/4)) \leqslant C|a-b|/\tau$. Average and use Cauchy–Schwarz to obtain the stated TV estimate. All conditioning in the proof was a construction under standalone Gaussian input. Disintegration retains the specified input kernel for subsequent uses; it does not turn the RMS estimate into a uniform conditional-on-input bound. □

The single quantitative certificate. The public block S_1^K supplies the dyadic heat decoder, and the retained mean supplies the normalized sampling root below. Section 13 pays the residuals, the scalar test, and the deterministic shift budget with internal mean cap $h^{-1/64}$. The external root has cap $h^{-1/32}$. Both public accuracy regimes use this one certificate, with all program choices fixed before oracle answers.

12.5 A normalized conditional-sampling root

Put $a = 2h^2$, $\ell = \sqrt{a}$, $s = 1/2$, $n = \sqrt{3}/2$, $F(z) = V(\sqrt{h}\,z)$, $x_F = x/\sqrt{h}$, and $r_F = \sqrt{2h}$. For $0 \leqslant \rho \leqslant 1/2$, let Y_u be the exact probability flow

$$\begin{aligned} \dot{Y}_u &= -r_F u_{E_u,r_u} = -\ell\nabla U_u(A_u), \\ E_u &= x_F + r_F u Y_u, \qquad r_u = r_F\sqrt{1-u^2}, \\ A_u &= x + \ell u Y_u, \qquad U_u = V_{a(1-u^2)}. \end{aligned}$$

Its marginal is $\xi_{x,u} = \mathsf{Law}(uZ + \sqrt{1-u^2}G)$, where $Z \sim \nu_{x_F,r_F}$ and $G \sim \gamma_d$ are independent.

Given the entering momentum P, make the fixed split

$$Y_0 = sP + nE_0, \qquad H_0 = nP - sE_0, \qquad E_0 \sim \gamma_d, \tag{12.18}$$

independently of the stopping endpoint. Let (u_i,ω_i) be the positive final quadrature rule on $[0,\rho]$, so $\sum_i \omega_i = \rho$. Define

$$c_i = -sr_F\omega_i, \qquad \sigma_f = \frac{n}{sr_F}, \qquad \beta = \sqrt{1-|\omega|^2} \geqslant \sqrt{3}/2.$$

The active mean inputs are sampled by the conditional Gaussian channel

$$(P_1,\dots,P_m) = -\omega \otimes H_0 + (I - \omega\omega^{\mathsf{T}})^{1/2}Z', \qquad Z' \sim \gamma_{md}. \tag{12.19}$$

Use the shared Gaussian iteration to supply the probability-flow centers, with standalone private mean inputs. Invoke fresh final means at precision K, radius r_{u_i}, width σ_f, and designated input P_i, transforming only their auxiliary tapes. Return

$$\widehat{W}_\rho(x,P) = P + \sum_i c_i \widehat{g}_i, \qquad \widehat{g}_i = \mathsf{M}^K_{\widehat{E}_i, r_{u_i}, \sigma_f}(P_i) - \sigma_f P_i. \tag{12.20}$$

Every coefficient is fixed before the gradient answers. No output residual is drawn, and no coefficient divides by the stopping endpoint.

Proposition 12.6 (The retained normalized sampling root)**.** *Under standard Gaussian input, the procedure has an input-retaining reference*

$$P = sY_0 + nH_0, \qquad W_\rho = sY_\rho + nH_1,$$

where, conditional on the exact parent path, (H_0, H_1) has standard Gaussian marginals. Thus $W_\rho \sim \xi_{x,s\rho}$. For $x \sim \pi_{V_a}$ and the schedule in section 13,

$$\|\widehat{W}_\rho - W_\rho\|_2 \leqslant C\sqrt{d}\, h^{K-1}, \qquad Q_{\mathrm{root}} \leqslant h^{-1/32}, \tag{12.21}$$

uniformly in ρ. The query cap holds at every center, for every fixed input and auxiliary tape. Its exact innovation satisfies

$$n^2 \mathbb{E}[\Delta H \Delta H^{\mathsf{T}} \mid \text{parent}] \preceq 2s^2a^2 \sum_i \omega_i^2 (1-u_i^2) \nabla^2 U_{u_i}(A_{u_i}). \tag{12.22}$$

Proof. For standard Gaussian P, (Y_0, H_0) is product Gaussian and does not depend on the stopping endpoint. The P_i are independent standard Gaussians, independent of Y_0. The comparison coordinate $G_E = \beta H_0 + \omega^{\mathsf{T}} Z'$ is an independent standard Gaussian, and $H_0 = -\sum_i \omega_i P_i + \beta G_E$. Apply the exact child kernels at the stationary centers, independently conditional on the parent path, to obtain P_i', and put

$$H_1 = -\sum_i \omega_i P_i' + \beta G_E.$$

The conditional product law gives the claimed Gaussian endpoints. Integrating the exact parent flow identifies $W_\rho = sY_\rho + nH_1$; replacing its integral by the final quadrature is the only remaining output defect. By (7.4),

$$n^2 \mathbb{E}[\Delta H \Delta H^{\mathsf{T}} \mid \text{parent}] \preceq \sum_i c_i^2 \operatorname{Cov}_{R^F_{r_{u_i}^2, E_{u_i}}}(\nabla F).$$

Rescaling and posterior Poincaré give (12.22).

For completeness, choose $J_r = \lceil h^{-1/128} \rceil$ uniform cells, $m_r = \lceil 128(K+4) \rceil$ GL nodes per cell, and $D_r = K+4$ synchronous layers. Cell lengths are at most $h^{1/128}/2$, and the center inventory is $N_r \leqslant C m_r J_r$. The physical rows are

$$b_i = x_F + r_F u_i Y_0, \qquad a_{ij} = -r_F^2 u_i w_{ij}, \qquad \max_i \sum_j |a_{ij}| \leqslant r_F^2/4.$$

Take $D_s = h^{-t}$, common center noise $\rho_c = r_F/(4D_s)$, and width $\sigma_c = 1/(2D_s\sqrt{N_r}\, r_F)$. The residual covariance is at least $3\rho_c^2 I/4$. All node radii are at least $\sqrt{3} r_F/2$, and

$$\lambda r_i^2 \leqslant 2h^2, \qquad \lambda r_i/\sigma_c \leqslant 4\sqrt{N_r}\, h^{2-t}, \qquad \lambda r_i/\sigma_f \leqslant Ch^2.$$

The schedule makes these at most h, so every call satisfies the mean-root hypotheses of proposition 12.5.

Under stationary x, Gaussian conditioning gives $(A_u, Y_u) \sim \pi_{U_u} \otimes \gamma_d$; hence the benchmark moments in (12.5) obey $\overline{D} \leqslant C\sqrt{d}$. On $[0, 1/2]$ the material derivative bounds have no endpoint singularity. Their order-m_r envelope is paid by section 13, and the quadrature defects are at most $C\sqrt{d}\, h^{K+3}$ in output units. The recurrence has contraction $2\lambda A \leqslant h^2$ and intrinsic source $C\rho_c h^K \sqrt{d}$. After D_r layers, its final center contribution is multiplied by $2\lambda \sum_i |c_i|$, while the active intrinsic error is at most

$$C\sigma_f \sum_i |c_i| h^K \overline{D} = Cn\rho h^K \overline{D} \leqslant C\sqrt{d}\, h^K.$$

Together with the output quadrature this proves the RMS bound. Auxiliary-only absorption retains every P_i and therefore the entering P.

Finally the internal mean cap is $h^{-1/64}$. The direct root inventory is at most $C(K+4)^2 h^{-1/128} \leqslant h^{-1/64}$ under the same cubic threshold. Multiplication gives the every-execution cap $h^{-1/32}$. This root needs no shift or Gaussian reserve of its own. □

13 One primitive and one error budget

We use the retained sampling-root interface of proposition 12.6 and the contraction estimate proposition 5.2 with rate $c\Gamma$. Each phase executes one such root. The approximation order $K \geqslant 10$ and resolution $h = e^{-y}$ determine all remaining choices. Throughout, the public curvature and warmness hypotheses hold, including $\kappa \leqslant d^{100}$.

One numerical certificate. Fix sufficiently small absolute $c_{\mathrm{sch}} > 0$ and sufficiently large absolute C_{sch}, before K, d, y and all oracle answers. Reserve the internal mean-work exponent $\gamma_M = 1/64$ and take

$$\begin{aligned} t &= \frac{c_{\mathrm{sch}}}{K+2}, \quad N_{\mathrm{P}} = \lceil C_{\mathrm{sch}}(K+2) \rceil, \quad D_0 = N_{\mathrm{P}} + 1, \\ H &= \left\lceil \frac{C_{\mathrm{sch}}(K+2)}{t} \right\rceil, \quad H_s = \lceil C_{\mathrm{sch}}(K+2) \rceil, \quad \mathcal{T} = \frac{D_0(H+1)}{\gamma_M}, \\ w &= \frac{c_{\mathrm{sch}}}{\mathcal{T}}, \quad m = \left\lceil \frac{C_{\mathrm{sch}}(K+2)}{w} \right\rceil, \qquad R = h^t, \quad \psi = h^{4t}, \quad h_0 = h^w. \end{aligned} \tag{13.1}$$

Thus $\mathcal{T} \asymp K^3$, $w \asymp K^{-3}$ and $m \asymp K^4$. All analytic and compiler requirements follow from one payment:

$$y \geqslant C_0 K^3 \ell, \qquad \ell = \log(K+2) + \log(1+y). \tag{13.2}$$

In particular this holds for every

$$y \geqslant C_1 K^3 \log(K+2). \tag{13.3}$$

Indeed, at $y_0 = C_1 K^3 \log(K+2)$ the bracket is at most $C[\log(K+2) + \log(1+C_1)]$, paid by sufficiently large C_1. The ratio $[\log(K+2) + \log(1+y)]/y$ decreases, so the same payment holds for every larger y.

The dimension-free factorial estimates place all derivative constants in one envelope with $\log A_{\mathrm{num}} \leqslant CK^4\ell$. Condition (13.2) pays $A_{\mathrm{num}} h^K \leqslant 1$. The raw simultaneous-iteration defect is

$$A_{\mathrm{num}} \{ h^{N_{\mathrm{P}}/3} + R^{-4} h_0^{m-1} \}.$$

Both residual powers are at least $2K + 20$, by the fixed schedule. Thus every internal numerical interface has error at most h^{K+1} in its original units, before recursive evaluation.

The full and active-support ranks are bounded by H and H_s. A common tier majorant gives

$$\text{work} \leqslant W := \mathcal{B}^{H+1}, \qquad \text{shift} \leqslant W^{\rho}, \quad \rho = \frac{H_s+1}{H+1},$$

where $\log W \leqslant CD_0(H+1)\ell + CwD_0(H+1)y$. First choose c_{sch} small and then C_0 large. This gives $W \leqslant h^{-1/64}$ and, since $\rho/64 \leqslant t$, the shift cap h^{-t}. Every private supplier and zero-shift subtree is included. The scalar envelopes satisfy $\log S \leqslant CH\ell$ and $\log B_0 \leqslant C\ell$, so the same payment gives their smallness test $CHB_0Sh^{1/4} \leqslant 1$ and root precision h^{K-1}.

The normalized sampling root has mesh at most $h^{1/128}$, order $\lceil 128(K+4)\rceil$, and $K+4$ simultaneous updates. Its inventory is at most $C(K+4)^2h^{-1/128} \leqslant h^{-1/64}$ under (13.3). Together with the internal mean cap, this gives its every-execution bound $Q_{\mathrm{pr}} \leqslant h^{-1/32}$ and stationary RMS error $C\sqrt{d}\,h^{K-1}$. These caps hold at every center, including inaccurate intermediate centers. Every order, seed layout, and cap is fixed before oracle answers.

One logarithm for phases and heat decoding. The latent reference law is $\mu = \pi_{V_{2h^2}} \otimes \gamma_d$, and $\Gamma = h/\sqrt{\kappa + 2h^2}$. A supplied sample warm for this joint law initializes the chain directly, with its momentum correlation preserved. An original-target warm sample is lifted by independent $\mathcal{N}(0, 2h^2 I)$ position noise and independent Gaussian momentum. These Gaussian channels preserve the declared squared L^2 bound B.

Execute N kinetic-heat phases, each using one normalized sampling root. Then execute L dyadic noisy-sampling updates, halving the heat variance each time, and return the final center. The retained-input Gaussian-reserve and cooling interfaces give stationary path entropy

$$Cd\{Nh^{2K-2}/\Gamma + Lh^{2K-2}\}. \tag{13.4}$$

Under stationary input, the exact decoder returns law $\pi_V * \mathcal{N}(0, 2h^2 2^{-L} I)$; its TV bias is at most $Ch\sqrt{d}\,2^{-L/2}$ by smoothness and Gaussian-mixture convexity.

Choose

$$\Lambda = \log \frac{CBd}{\epsilon^2}, \qquad N = \lceil C\Gamma^{-1}\Lambda\rceil, \qquad L = \lceil C\Lambda\rceil. \tag{13.5}$$

Then $L \leqslant CN$ and $N \leqslant C\sqrt{\kappa}h^{-1}\Lambda$. Exact mixing costs at most $\epsilon/4$. Apply conditional Pinsker, then Cauchy–Schwarz against the initial relative density, to the stationary path entropy. The cooling term in (13.4) is absorbed into the phase term, giving

$$\mathsf{TV}(\mathsf{Law}(A_{\mathrm{out}}), \pi_V) \leqslant \epsilon/4 + C\sqrt{Bd\kappa\Lambda}\,h^{K-2} + Ch\sqrt{d}\,2^{-L/2}. \tag{13.6}$$

The chosen L pays the heat-decoding bias by $\epsilon/4$. Every execution uses at most

$$Q \leqslant C(N+L)h^{-1/32} \leqslant C\sqrt{\kappa}\,\Lambda e^{(33/32)y}. \tag{13.7}$$

The stated fixed-resolution window. Set $B = d^{C_w}$, $S = (C_w+1)/2$, $y = e\log d$, and retain $K = \lceil 10 + (101+S)/e\rceil$. The stated $\log d \geqslant X_3 = (C_{\mathrm{th}}/e)K^3\log(K+2)$ implies (13.3) for $C_{\mathrm{th}} \geqslant C_1$. For $\epsilon \geqslant d^{-1}$, $\Lambda \leqslant CK(1+y/e)$; because $1/e \leqslant K$, its logarithm is at most $C[\log(K+2) + \log(1+y)]$. Furthermore

$$\sqrt{Bd\kappa}\,h^{K-2} \leqslant d^{S+50-e(K-2)} \leqslant d^{-51-8e}.$$

The numerical payment absorbs $C\sqrt{\Lambda}$, giving the required accuracy. It also pays $\log(C\Lambda)$ by $y/2$ in (13.7), so $Q \leqslant C\sqrt{\kappa}d^{49e/32} \leqslant C\sqrt{\kappa}d^{2e}$. For theorem 1.1, take $e = \min\{\eta/4, 1/16\}$ and increase d_0 so that $\log d_0 \geqslant X_3$ and $d_0 \geqslant \epsilon^{-1}$. This proves the fixed-accuracy result with the same schedule and the supplied joint warm input.

Arbitrary accuracy in the original dimension. For an original-target warm input set

$$\mathcal{L} = 2 + \log \frac{\sqrt{Bd\kappa}}{\epsilon}, \qquad y = C_* \left\{ K^3 \log(K+2) + \frac{\mathcal{L}}{K} \right\}.$$

The first term pays (13.3) for every $\mathcal{L}$. Since $\Lambda \leqslant C\mathcal{L}$ and $K - 2 \geqslant K/2$, the second term pays (13.6). The inequality $\log \mathcal{L} \leqslant \log(K+2) + \mathcal{L}/K$ pays the coefficient in (13.7) by $y/2$, after an absolute enlargement of C_*. Thus $Q \leqslant C\sqrt{\kappa} e^{2y}$. Choose $K = \lceil 10 + (\mathcal{L}/\log \mathcal{L})^{1/4} \rceil$. Then $y = O(\mathcal{L}^{3/4} (\log \mathcal{L})^{1/4})$. Under $B \leqslant d^{C_w}$ and $\kappa \leqslant d^{100}$, $\mathcal{L} \leqslant C_{C_w} L_0$, proving (1.1) directly in dimension d. The universal threshold constant is existential, as in the stated theorems.

Source correspondence and attribution

Analytic source and contribution boundary. The analytic conditional-mean, probability-transport, Stein-centering, finite numerical, and recursive original-gradient constructions adapt OpenAI, *Subpolynomial query complexity for well-conditioned log-concave sampling* [Ope26], at the pinned repository version specified in the bibliography. The inherited statements, constructions, and proofs are not claimed as new results of this manuscript. OpenAI is the immediate source of the adapted exposition, not necessarily the original source of every standard mathematical ingredient.

The correspondence below identifies constructions rather than asserting verbatim identity of statements or proofs. It uses section numbers in the pinned OpenAI source and the labels of the present revision.

Present revision	OpenAI source	Inherited construction
Sections 6 and 8	Sections 2–3	Conditional means, Gaussian conditional estimates, and probability transport
Section 9	Section 5	Transported Stein centering and stationary mean representation
Section 10	Section 4	Adjoint-divergence and material-derivative framework; here reorganized into a second-moment calculus
Section 11	Section 6	Finite flow evaluation and conditional-mean numerical formulas
Section 12	Section 7	Gaussian absorption and recursive original-gradient evaluation

The retained-input covariance estimates, simultaneous affine evaluation, shared-layer implementation, and kinetic heat-cycle argument are the refinements organized here. The underlying noisy-mean machinery and Gaussian absorption are inherited, rather than independent inventions. Gaussian conditioning and reverse heat are credited to the proximal-sampling literature [CCSW22, KV26]; the present Poisson and Bochner mixing toolkit is credited in particular to [Che26, CCLZ26a].

Relation to the earlier public draft. The earlier public draft, preserved in the original release, contains a statement-level source-correspondence table and earlier-source references in its Appendix M. Its latent Hamiltonian and gap-shifted corrector argument credited Fan, Li, and Lu. Their revised analysis [FLL26] also covers the unshifted corrector used in the present section 5. The present revision changes the algorithm and proof organization; the old statement numbering and exact proof correspondences should not be read as those of this revision. The current contribution is still a supplied-warm, exact-real original-gradient query guarantee; it is not a warm-start construction, an arithmetic-runtime bound, or a finite-bit implementation guarantee.

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Lean formalization and workflow acknowledgment. The accompanying Lean development addresses theorem 1.1 and corollary 1.2; theorem 1.3 is not included in that formalization. The development should not be read as formal verification of every result in this manuscript or as an extracted executable sampler. The accompanying release materials should be consulted for the precise theorem mapping, dependencies, and axiom audit. The formalization workflow drew on Scott Armstrong's *LeanAutoformalizationSkills* (revision `05a0731`), particularly its guidance on project architecture, library search, and proof-dependency planning. The collection's original material is copyright 2026 Scott Armstrong and licensed under Creative Commons Attribution 4.0 International. This acknowledgment concerns workflow guidance, not a Lean library dependency, authorship of the mathematical arguments, or endorsement.

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